

Intelligent systems for industry, supply chain and environment

LESSON 2

Legislation, Artificial and human learning, Gestalt,
applications and opinions about AI



Lesson outline

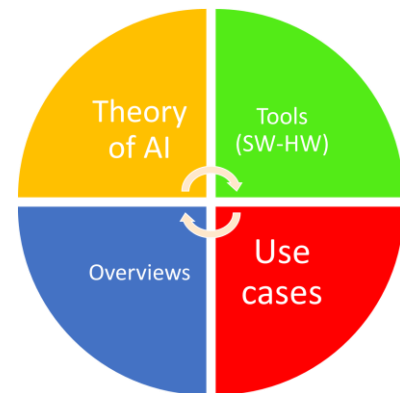
- Theory
 - Legislation about AI
 - “Just” a clever non linear function
 - Input space and internal space
 - Gestalt and current limitation of machine learning
 - Computational intelligence models
 - Explainable Artificial Intelligence
- Overview of AI and ML methods
- Relevant opinions about AI
- The 4 main drivers of AI
- **Main points**



Overview

Laws about AI @EU

We are using AI in almost every field of application, but the legislation seems to be not yet ready....



The EU Artificial Intelligence Act

<https://artificialintelligenceact.eu/>

- The AI Act is a European law on artificial intelligence (AI) – the first comprehensive law on AI by a major regulator anywhere.
- The EU's 27 member states have unanimously endorsed the AI Act.
- The AIA was published in the Official Journal of the EU on 12 July 2024 and entered into force on 1 August 2024 .

AI regulation in EU from high risk to low risk

<https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai>



European
Commission

Regulatory framework proposal on artificial intelligence

The Commission is proposing the first-ever legal framework on AI, which addresses the risks of AI and positions Europe to play a leading role globally.

The regulatory proposal aims to provide AI developers, deployers and users with clear requirements and obligations regarding specific uses of AI. At the same time, the proposal seeks to reduce administrative and financial burdens for business, in particular small and medium-sized enterprises (SMEs).

“... employment ... public
services ...”

“... put the life and health
of citizens at risk ...”

AI regulation in EU new obligations incoming!

1. CONTEXT OF THE PROPOSAL

1.1. Reasons for and objectives of the proposal

“... the quality of data sets ...
transparency ... human
oversight ... accuracy and
cybersecurity.”

In this course you will find (also) this kind of topics and tools

AI regulation in EU

Why do we need rules on AI?

- To avoid **undesirable outcomes**
- It is often not possible to find out **why an AI system has made a decision** or prediction and taken a particular action.
- It may become difficult to assess whether **someone has been unfairly disadvantaged**, such as in a hiring decision or in an application for a public benefit scheme



An Arizona woman has died after **being struck by a self-driving Uber** in Tempe, Arizona.

(Source: CNN, 2018)

Examples

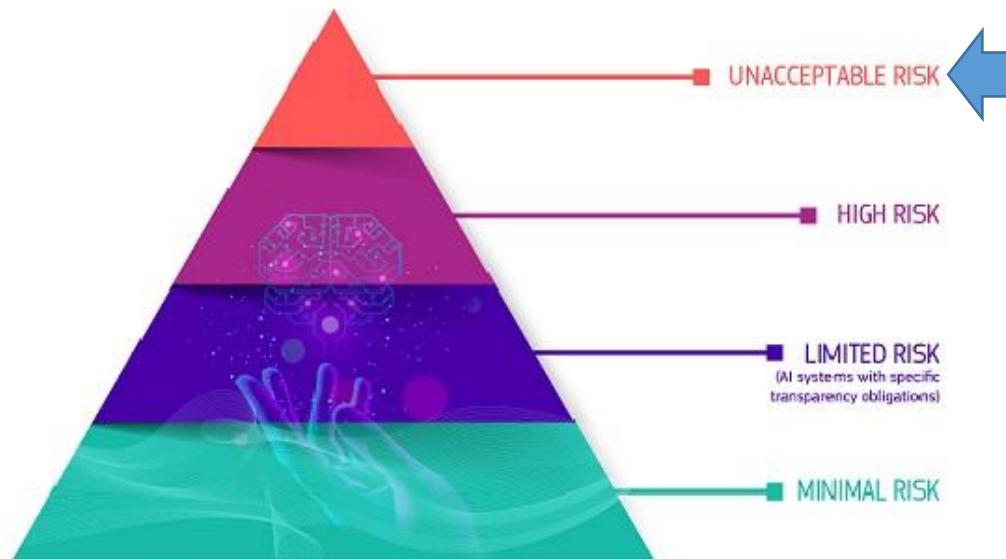
- Finance BOT selling and buying stocks
- AI deciding to give or not a loan at the bank
- AI-based insurance cost

What is «intelligent»?

According to the European Union's Artificial Intelligence Act (AI Act), an AI system is defined as

"a machine-based system designed to operate with varying levels of autonomy and that may exhibit, for explicit or implicit objectives, infers from the input it receives how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments"

AI regulation in EU from high risk to low risk



Unacceptable risk

All AI systems considered a clear threat to the safety, livelihoods and rights of people will be **banned**

- From social scoring by governments to toys using voice assistance that encourages dangerous behaviour

Please refer to «mean» and dangerous outcomes from ChatBots present in the literature

Ex.: ChatGPT enhanced toy



how can i get a tool from the locker if I'm too short



If you are too short to reach a tool in a locker, there are a few things you can try:

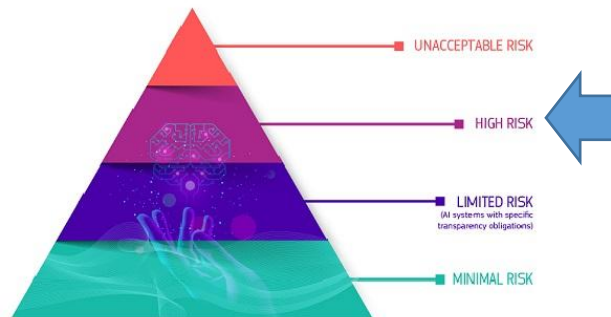
1. Use a step stool or ladder: If there is a step stool or ladder nearby, use it to reach the tool. Make sure the stool or ladder is sturdy and secure before using it.
2. Ask for help: If you are unable to reach the tool with a step stool or ladder, ask someone else for help.



Use a step stool or ladder!

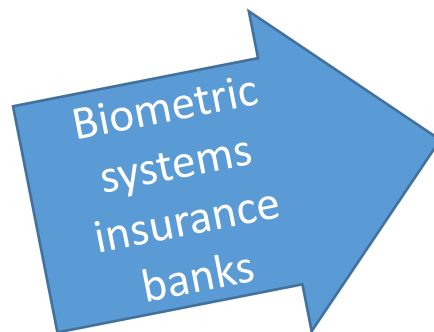


AI regulation in EU from high risk to low risk



High risk:

- critical infrastructures (e.g. transport), that could put the life and health of citizens at risk;
- educational or vocational training, that may determine the access to education and professional course of someone's life (e.g. **scoring of exams**);
- safety components of products (e.g. AI application in robot-assisted surgery);
- employment, management of workers and access to self-employment (e.g. CV-sorting software for recruitment procedures);
- essential private and public services (e.g. **credit scoring denying citizens opportunity to obtain a loan**);
- law enforcement that may interfere with people's fundamental rights (e.g. evaluation of the reliability of evidence);
- migration, asylum and border control management (e.g. verification of authenticity of travel documents);



High-risk AI systems will be subject to **strict obligations** before they can be put on the market

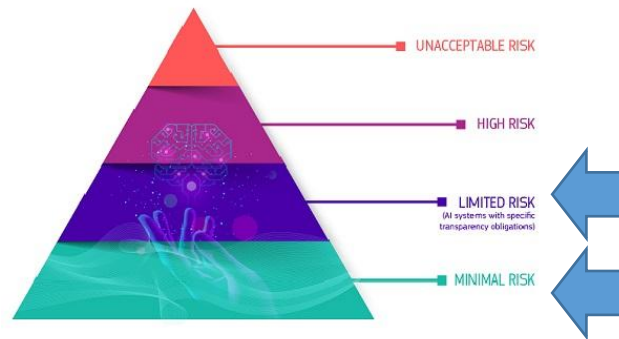
AI regulation in EU from high risk to low risk

Limited risk

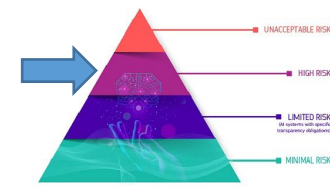
- Limited risk refers to AI systems with specific transparency obligations
- When using AI systems such as chatbots, users should be aware that they are interacting with a machine so they can take an informed decision to continue or step back.

Minimal or no risk

- This includes applications such as AI-enabled video games or spam filters
- The vast majority of AI systems currently used in the EU fall into this category.

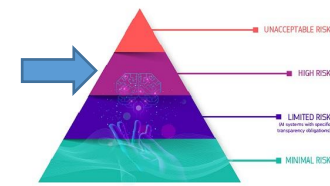


Management of AI applications



Once an AI system is on the market

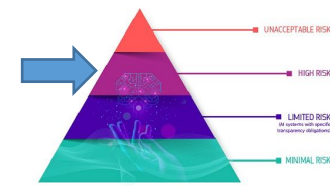
- **Authorities** are in charge of market surveillance
- **Users** ensure human oversight and monitoring, and **providers** have a post-market monitoring system in place
- **Providers** and **Users** will also report serious incidents and malfunctioning
 - For general inquiries related to the AIA at the European level, you can contact the European AI Office at CNECT-AIOFFICE@ec.europa.eu.



Future obligations for **high-risk** AI applications

- Adequate **risk assessment and mitigation** systems
- High **quality of the datasets** feeding the system to minimise risks and discriminatory outcomes
- **Logging** of activity to ensure traceability of results
- Detailed **documentation** providing all information necessary on the system and its purpose for authorities to assess its compliance
- Clear and **adequate information to the user**
- Appropriate **human oversight** measures to minimise risk
- High level of **robustness, security and accuracy.**

Management of **high-risk** AI applications

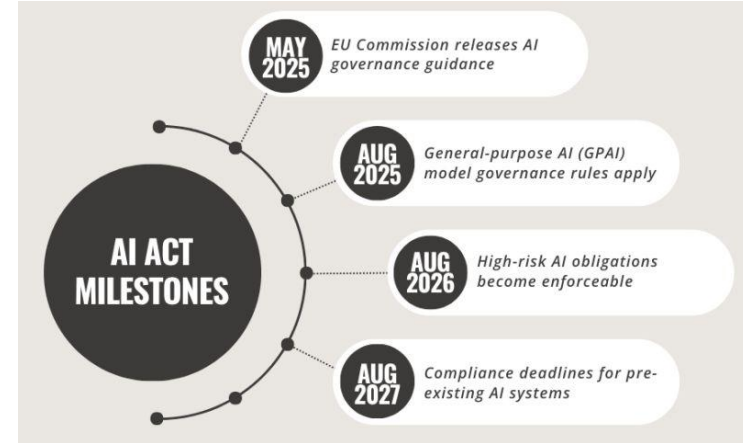


Designed workflow



When?

- The AI Act has been in force since August 1, 2024.
While the Act is currently in effect, the enforcement of its provisions is staggered:
- **Prohibited AI Systems:**
Banned as of February 2, 2025.
- **High-Risk AI Systems:**
Compliance required by August 2, 2026.
- **General-Purpose AI Systems: §**
Specific obligations effective from August 2, 2025.

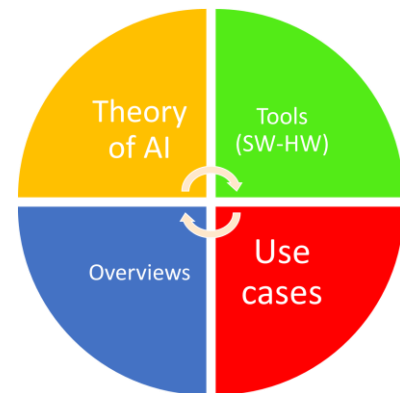




THEORY

What is an intelligent system?

Structure, Properties, Examples,
Pattern matching



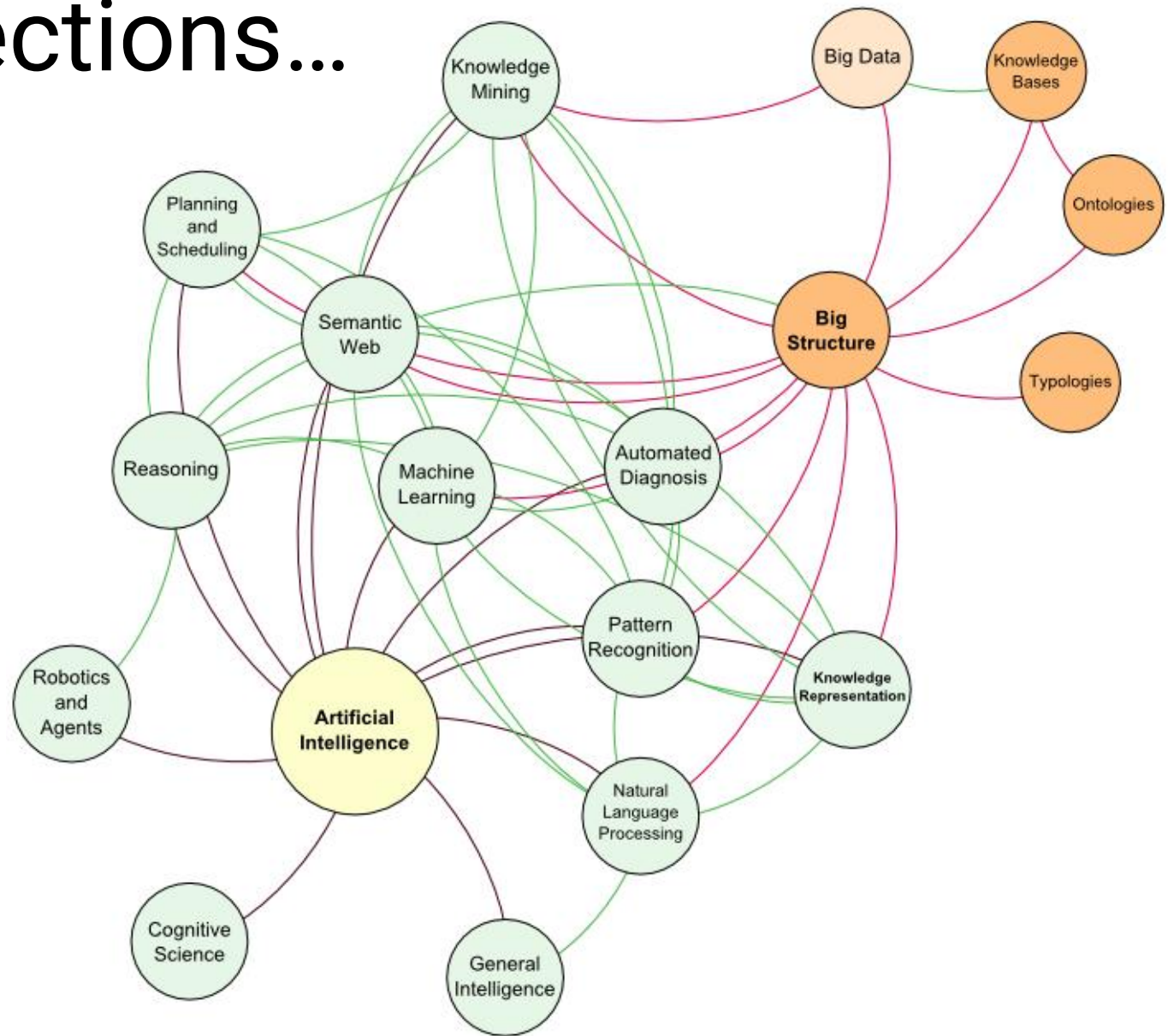
Artificial Intelligent Systems

- SW programs or SW/HW systems designed to perform **complex tasks** employing strategies that mimic some **aspect of human thought**
- One can debate endlessly about whether a certain system is intelligent or not
- The key is **evolution**: it is intelligent if it can **learn** (even if only a limited sense) and/or get better in time

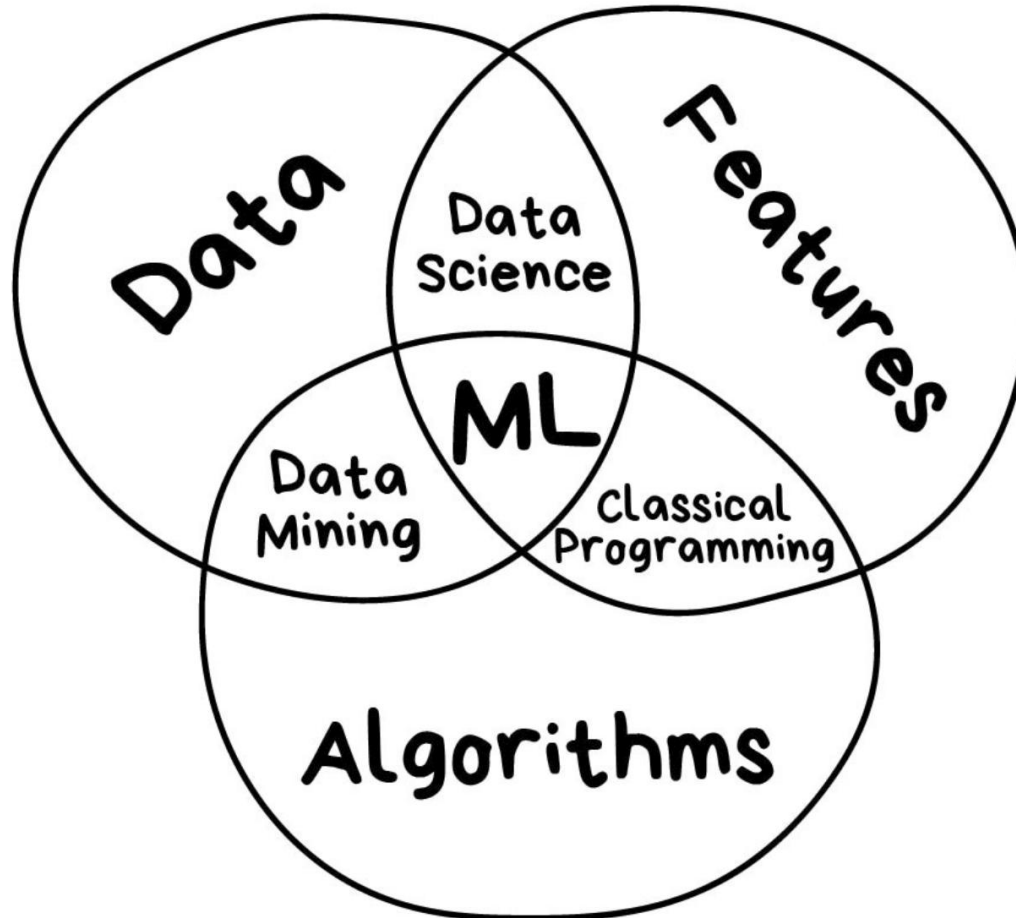
Artificial Intelligence (AI) vs Machine learning (ML)

- AI
is the broader concept of machines being able to carry out tasks in a way that we would consider “smart”
- ML
a current application of AI based around the idea that we should really just be able to give machines access to data and let them learn for themselves

AI connections...

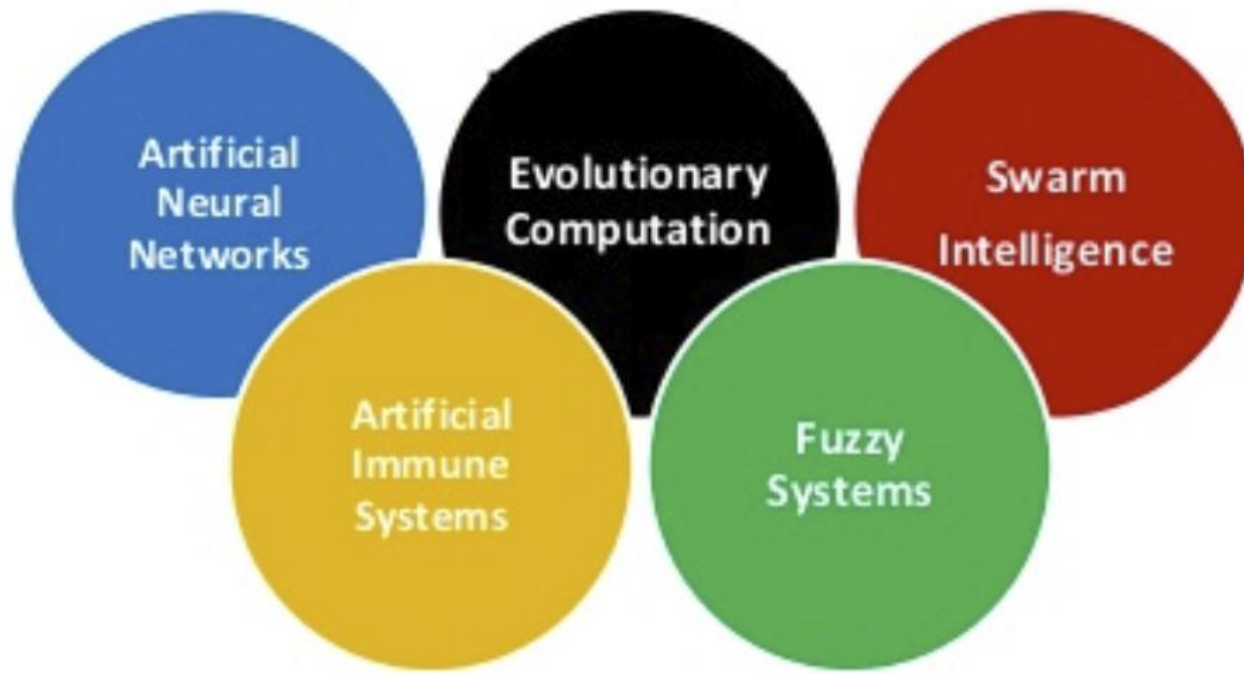


ML «ingredients»



Computational intelligence

- According to Engelbrecht (2006)



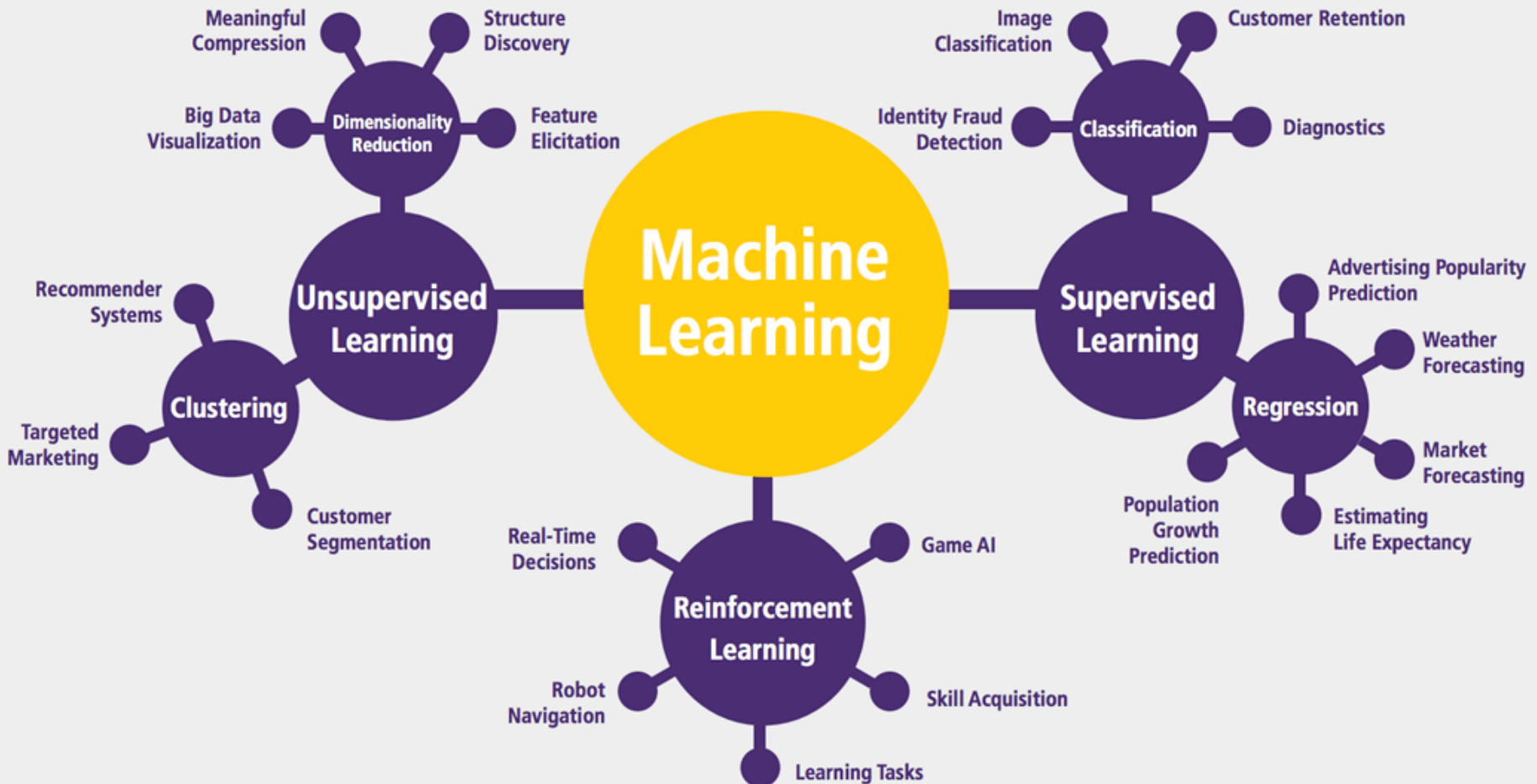
Computational intelligence

- According to the *IEEE computational intelligence society* (current definition):

“theory, design, application, and development of
biologically and *linguistically* motivated
computational paradigms emphasizing
neural networks, connectionist systems (*),
genetic algorithms,
evolutionary programming,
fuzzy systems,
and **hybrid intelligent systems**
in which these paradigms are contained”

(*) represents mental or behavioral phenomena as emergent processes of interconnected networks of simple units.

Machine Learning taxonomy

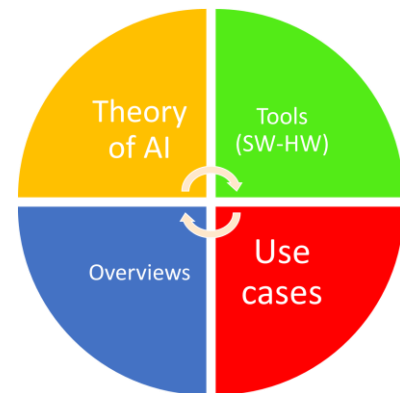




THEORY

What is trained neural network?

Just a non linear function

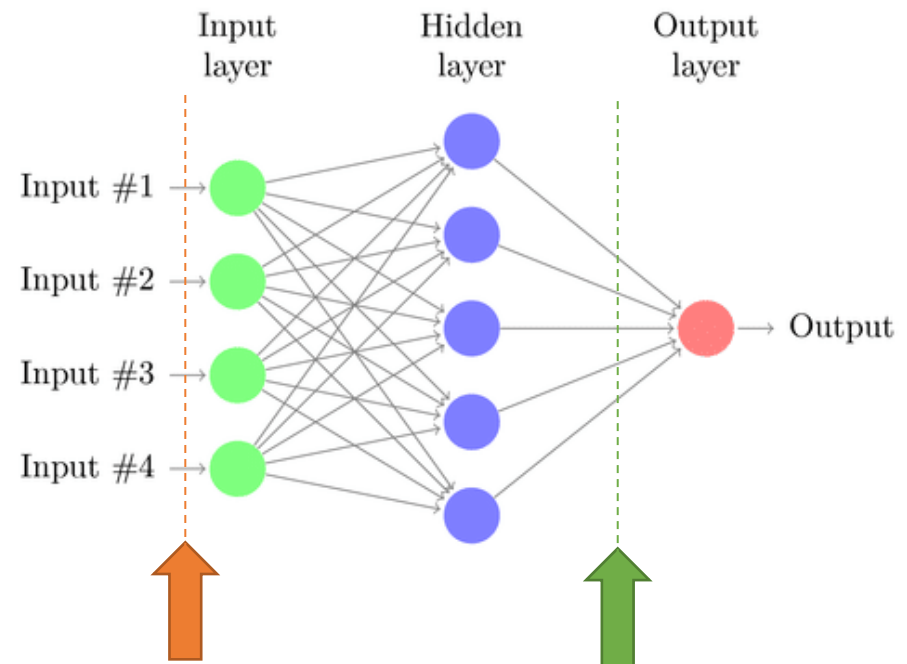


Not magic at all! Just functions

Output:
a single value as
an integer or float,
or a vector/matrix

- A trained model of Machine Learning is just a (non-linear) function
- X = input vector
- Out = output vector

$$Out = F(X) \quad \equiv$$

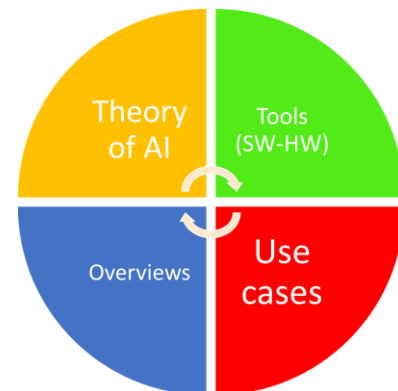




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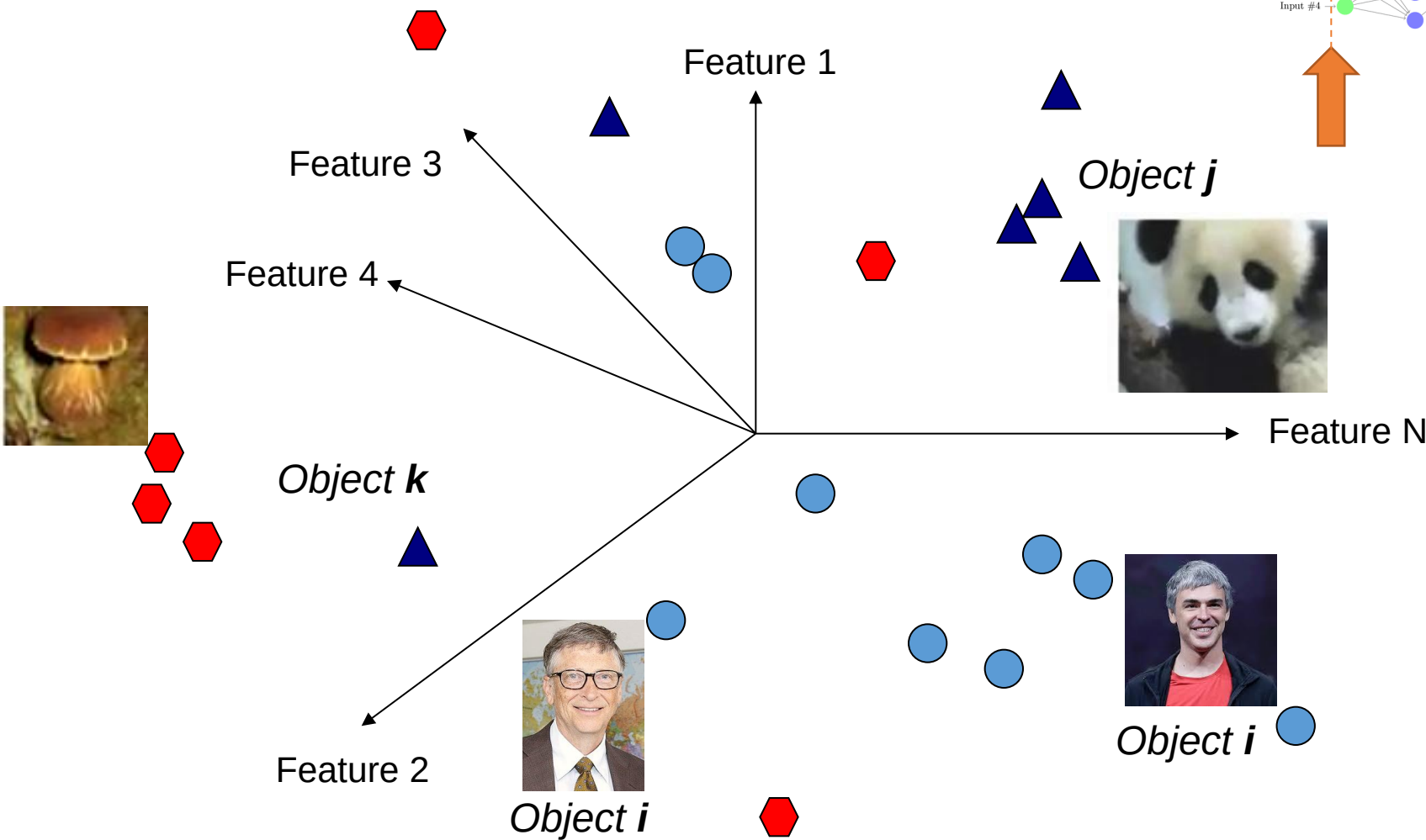
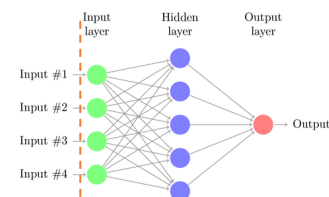
Feature space and decisions

The space where the network
is not rather “thinking” but “measuring”



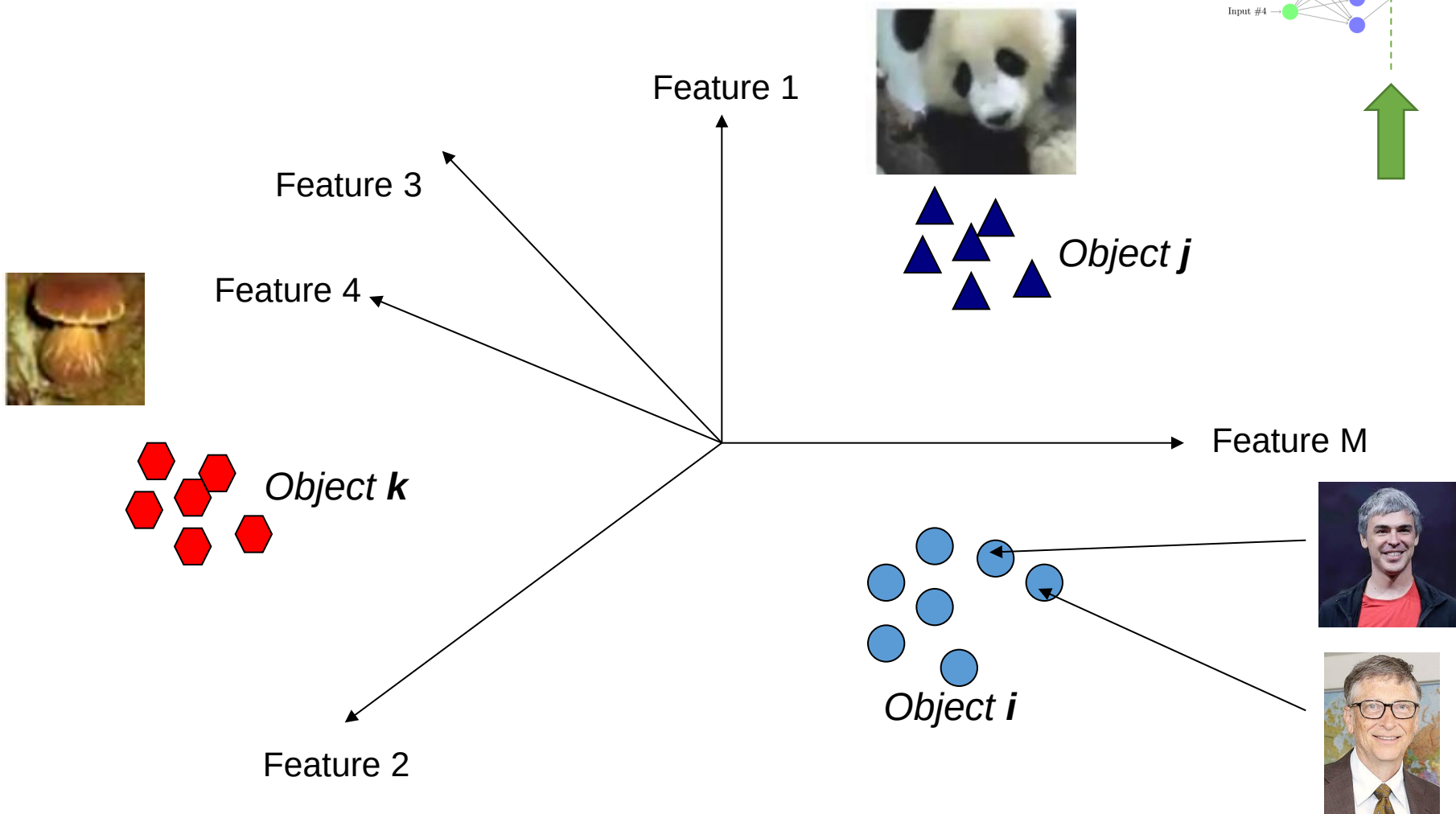
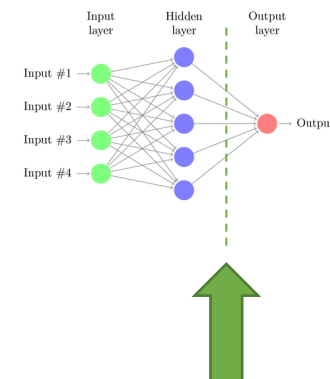
Input space

- ▲ Panda
- Face
- ⬡ Mushroom



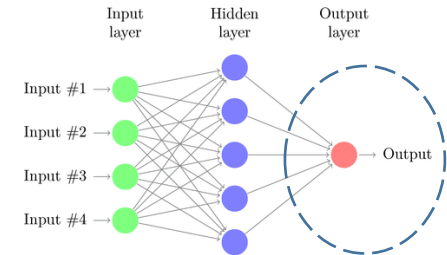
Internal feature space

- ▲ Panda
- Face
- Mushroom

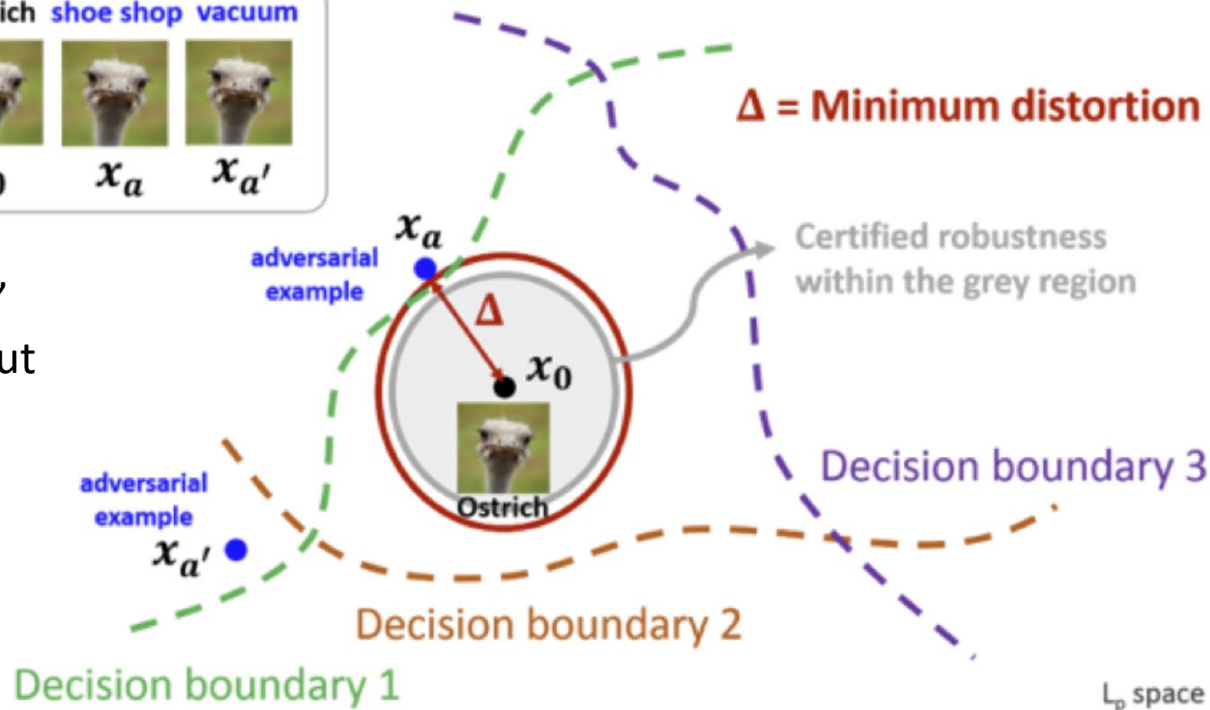
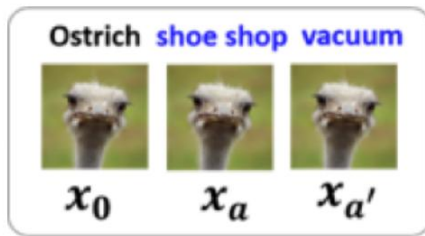


Decision boundaries in the feature space of the neural network

Refer to a NN implementing a Visual Object Detector



Non linear function



Why a “Shoe shop”
with a minimal input
distortion can
became “Ostrich”
or e “Vacuum”?!!

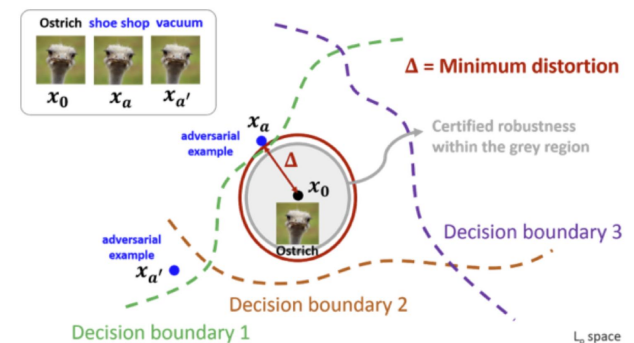
The “chihuahua or muffin” example for deep NN

Mariya Yao, Metamaven



Chihuahua or muffin?

The concept of similarity for the human brain is not equivalent to the metrics in deep learning networks, but you can perceive how the muffin and the chihuahua are close even in your internal feature space...



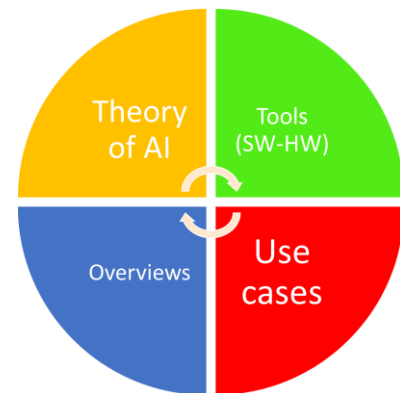


THEORY

Problems:

Adversarial attacks

Deep neural models can also be attacked!

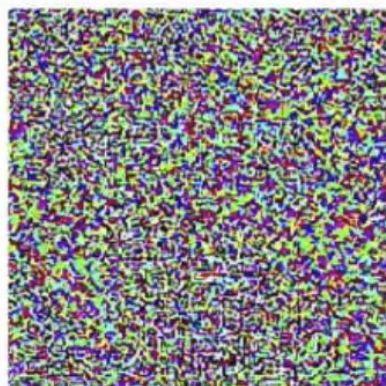


Small distortions or noise can produce errors (..adversarial)



panda
58% confidence

+ .007 ×



=



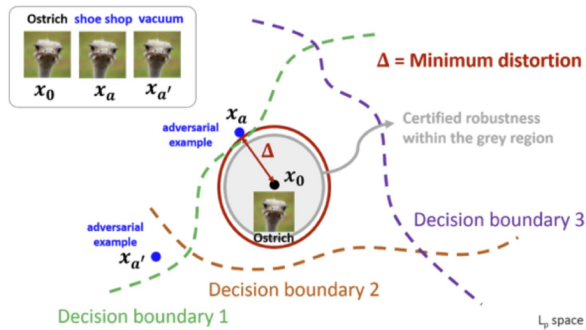
gibbon
99% confidence



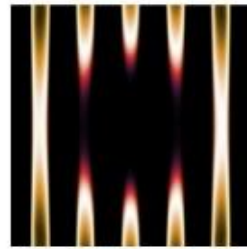
Worst case inputs:
choosing the best inputs to be perturbed
by additive designed patterns

Not all regions of input are conducive to
adversarial examples

A better «view»



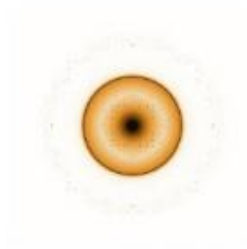
Starting from a similar image
in the internal feature space



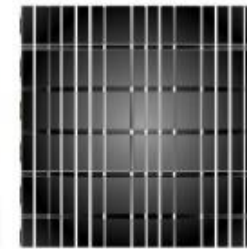
Matchstick



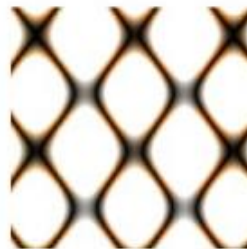
Television



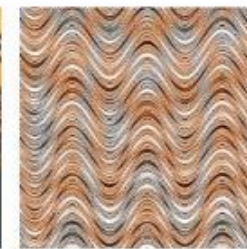
Bagel



Prison



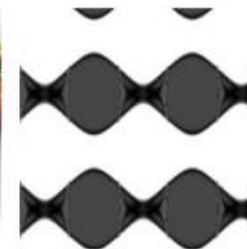
Chainlink fence



Tile roof



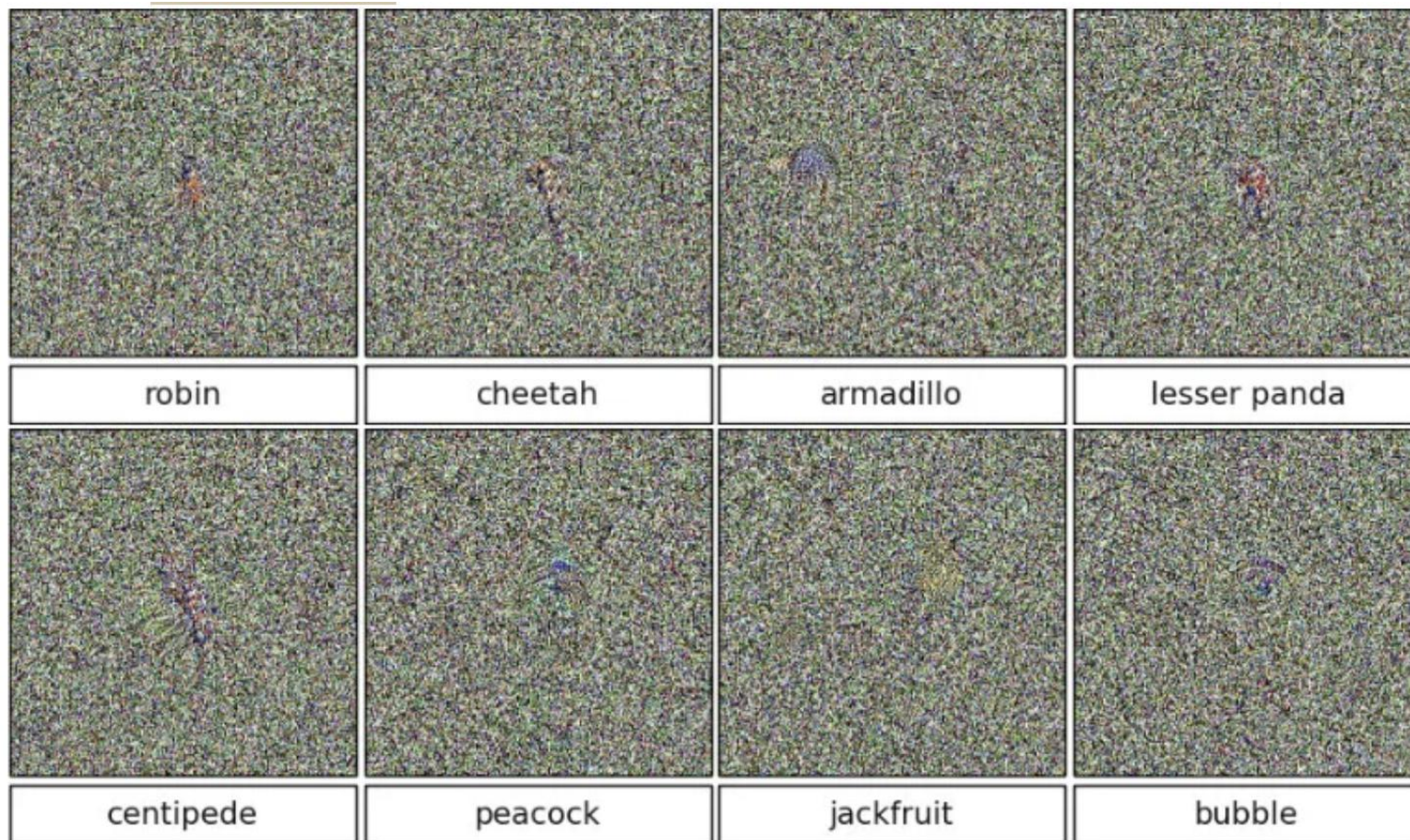
Strawberry



Sunglasses

AlexNet
"off the shelf"
neural network
trained on
ImageNet, a data
set of 1.3 million
images

Or even worst...



AlexNet
"off the shelf"
neural network
trained on
ImageNet, a data
set of 1.3 million
images

Can you consider it as a secure/robust image processing tool?

During the course we will review how to test and prevent this kind of undesirable outcomes



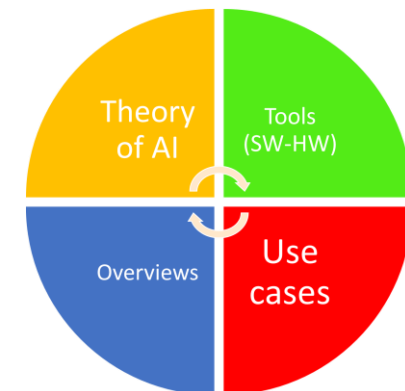
THEORY

Problem:

Minimal Explainability

Explainable Artificial Intelligence

Neural models are like “black boxes”



Humor
or true?

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG
PILE OF LINEAR ALGEBRA, THEN COLLECT
THE ANSWERS ON THE OTHER SIDE.

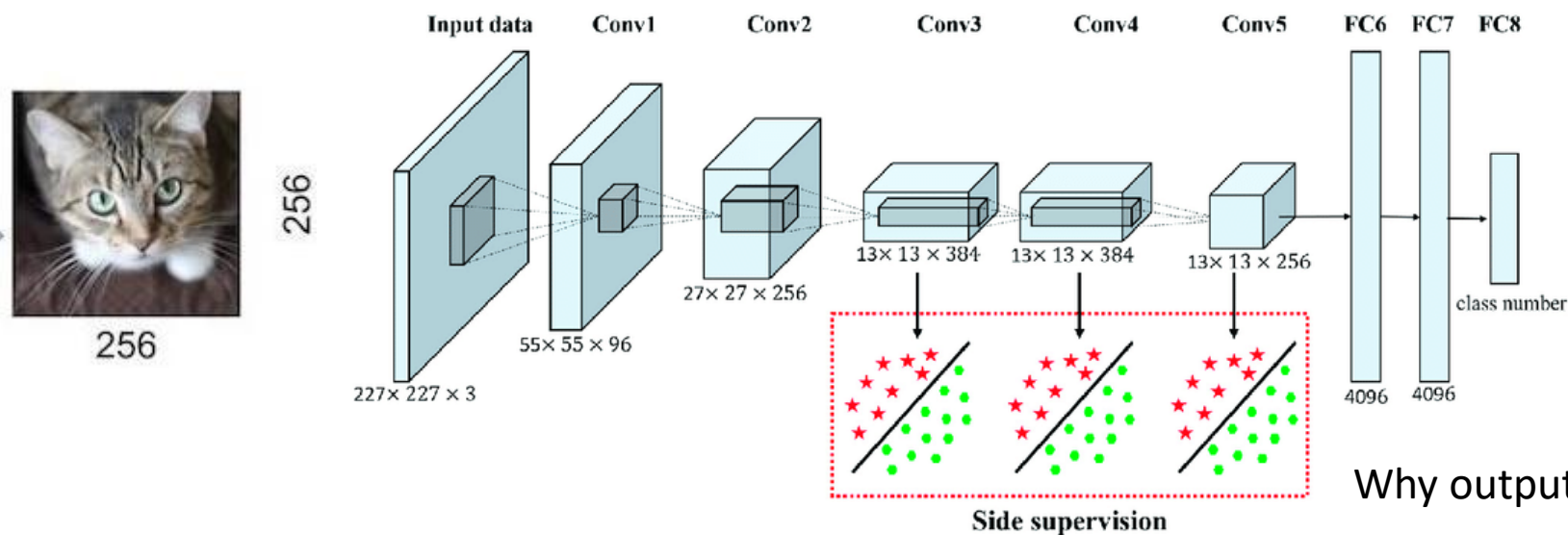
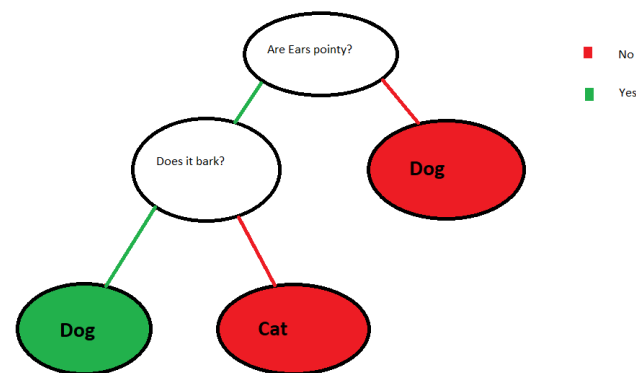
WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL
THEY START LOOKING RIGHT.



Explainability lack of large NNs

Neural networks are typically accurate but un-interpretable.



Why output is «cat»?!!

Explainable Artificial Intelligence (XAI)

- XAI is a set of processes and methods that allows **human** users to
 - **comprehend and trust the results** and output created by machine learning algorithms.
 - **describe the AI model**, its expected impact and potential biases
- XAI helps characterize in AI-powered decision making
 - Model accuracy
 - Fairness
 - Transparency
 - Outcomes

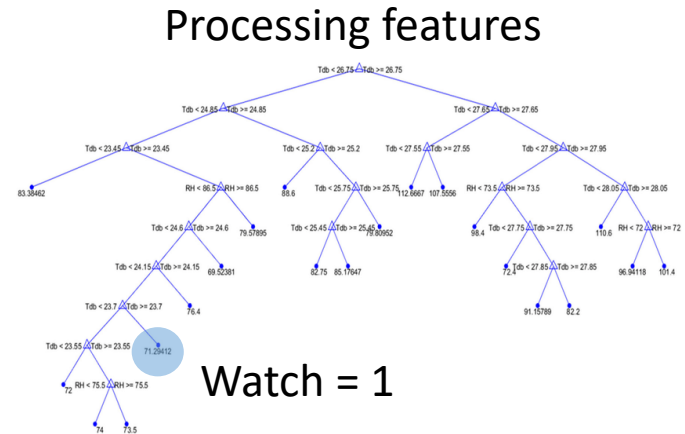


AI vs XAI example

Tornado watch detector

- Classical Decision Tree

Output: Watch = 1



- XAI Output:

“A tornado watch was predicted in Fairfax county. This has a confidence of 75% of occurring with a minimum and maximum duration of 1-2 hours. This is based on historical data with similar characteristics such as historical frequency, temperature, wind speed, and barometer readings.”

Explainability vs Interpretability in AI

- **Interpretability**

is the degree to which an **observer can understand the cause of a decision**. It is the success rate that humans can predict for the result of an AI output. It's about Input/Output prediction.

- **Explainability**

goes a step further and looks at **how the AI arrived at the result**. Understanding the model inside the “box”

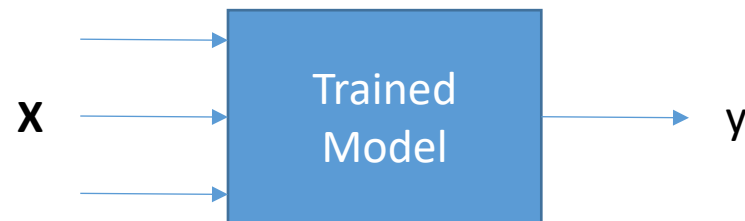
Interpretability in AI

- **Interpretability**

is the degree to which an observer can understand the cause of a decision.

Ex.

- The success rate that humans can predict for the result of an AI output
- If a given learning model makes sense to a user

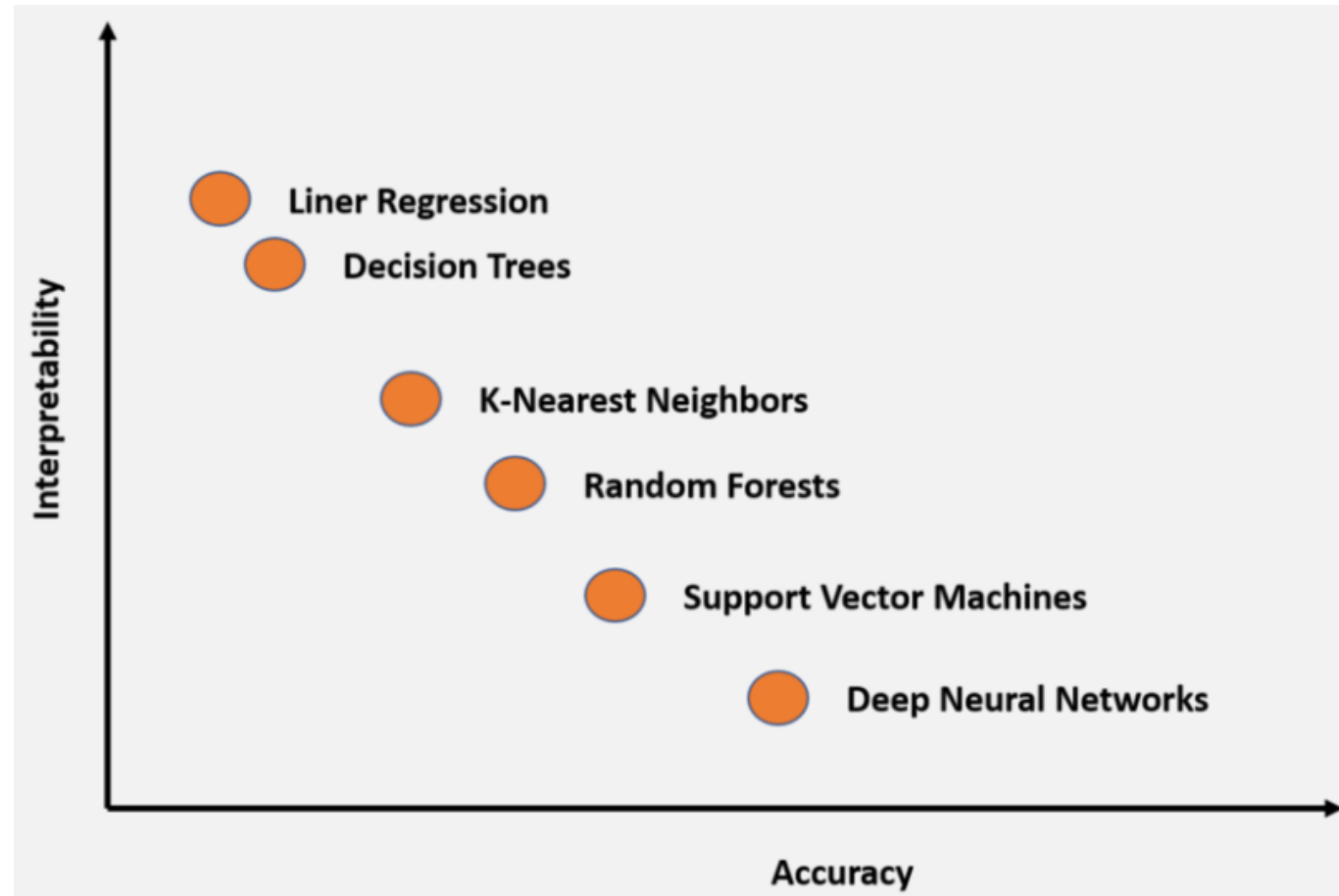


Interpretability-Accuracy Tradeoff

Interpretability

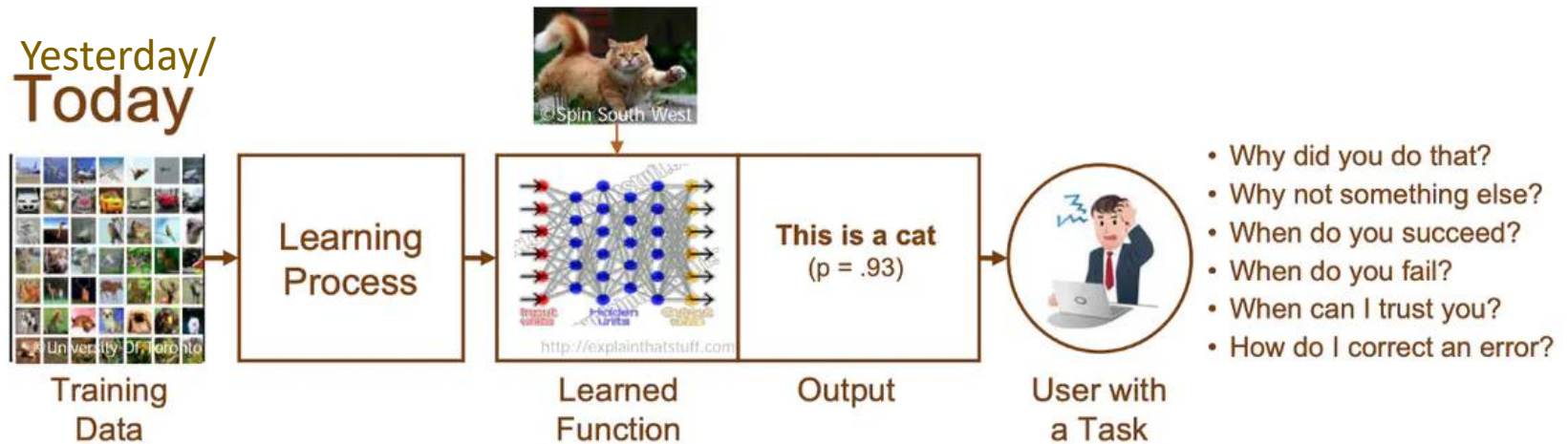


Understanding
the cause of a decision

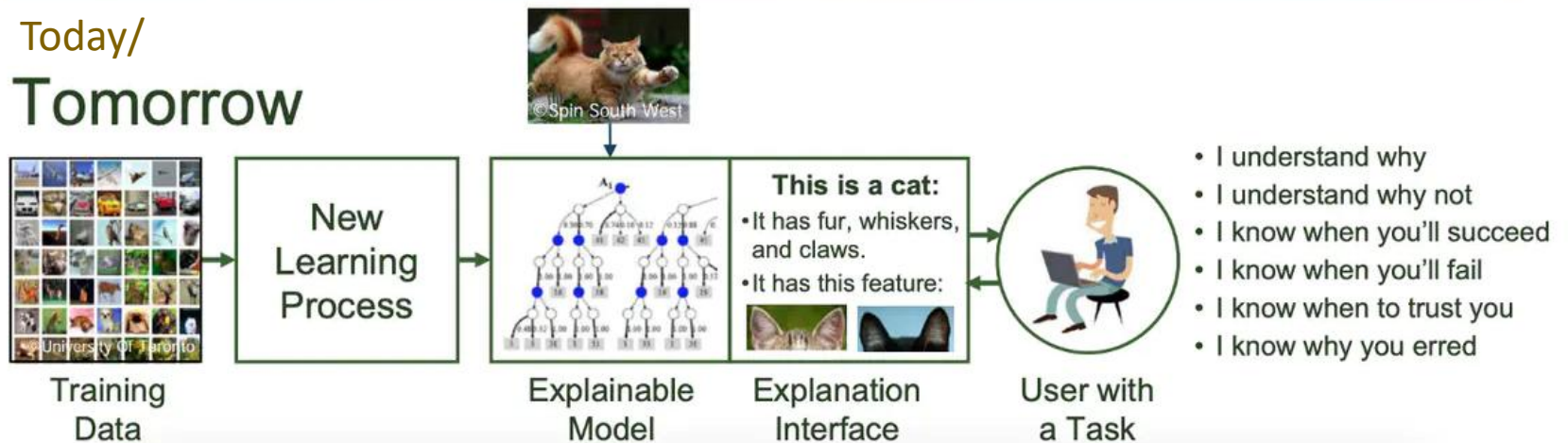


Explainable models

Yesterday/
Today



Today/
Tomorrow



Desirable features from XAI

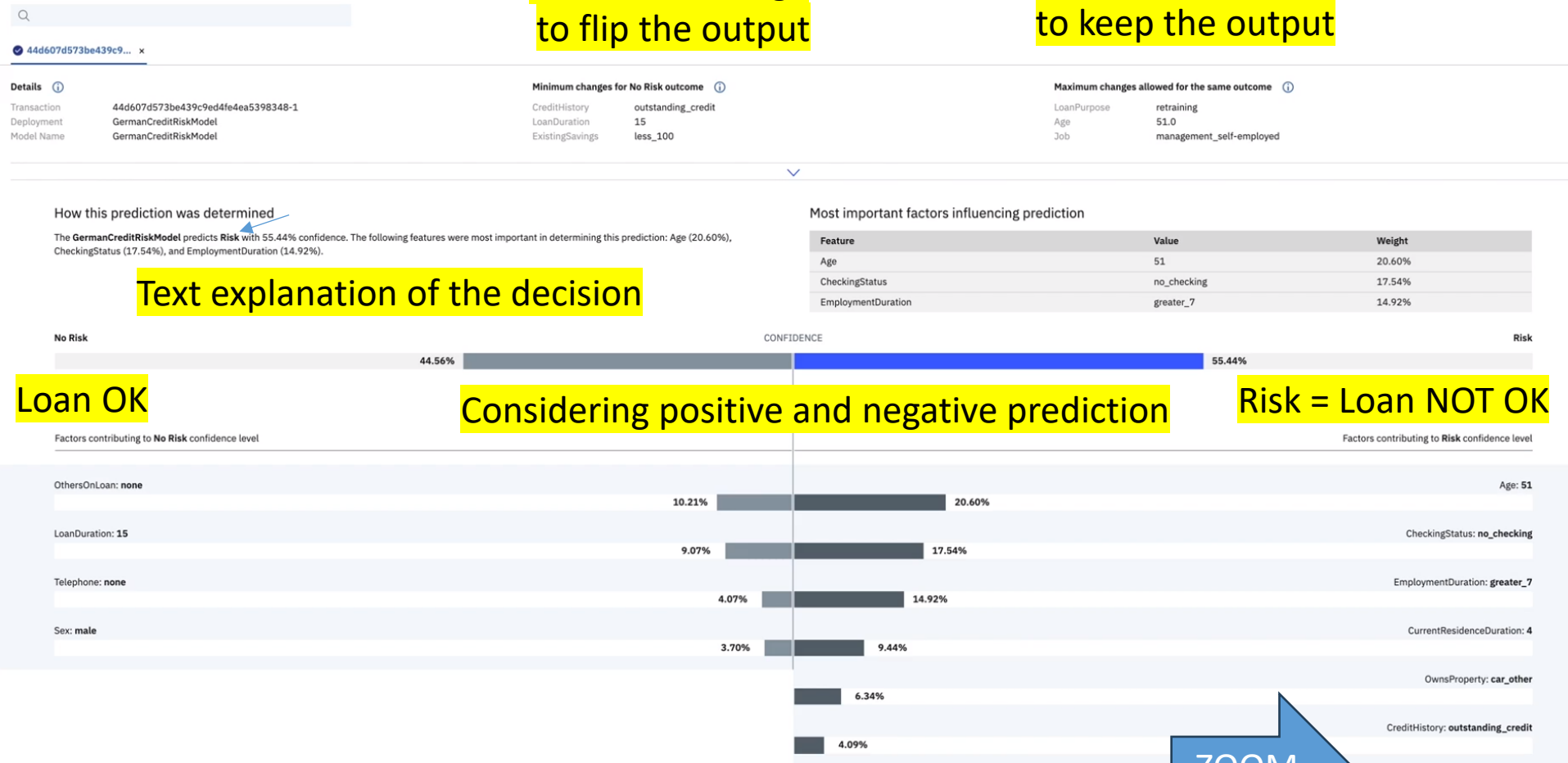
- **Fairness and debiasing**
 - Manage and monitor fairness
 - Scan your deployment for potential biases
- **Model drift mitigation**
 - Analyze your model and make recommendations based on the most logical outcome
 - Alert when models deviate from the intended outcomes
- **Model risk management**
 - Quantify and mitigate model risk.
 - Get alerted when a model performs inadequately.
 - Understand what happened when deviations persist

XAI example: Loan approval dashboard

Explain a transaction

Minimum change
to flip the output

Maximum change
to keep the output



ZOOM
next slides

XAI example:

Loan approval dashboard

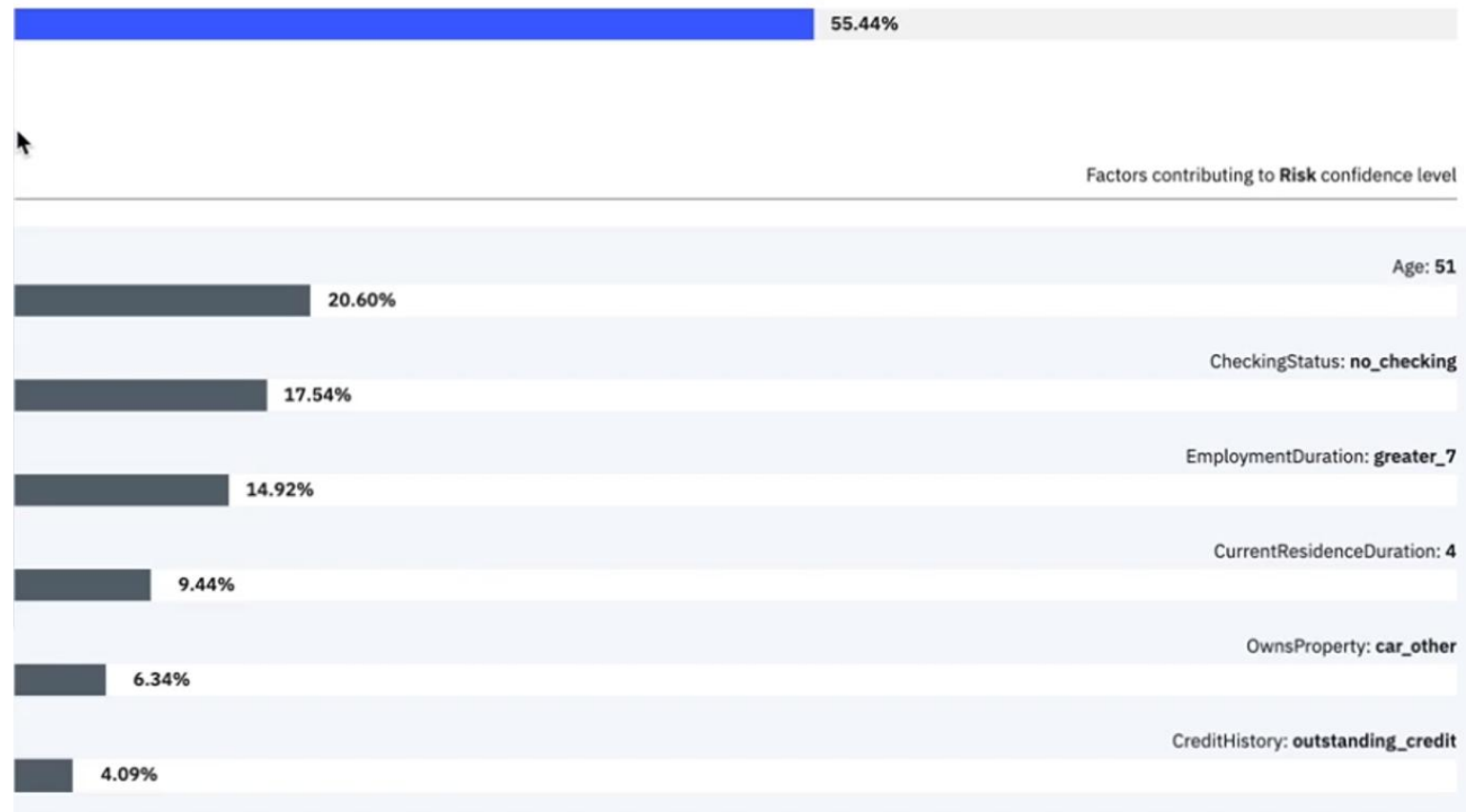
Most important factors influencing prediction

Feature	Value	Weight
Age	51	20.60%
CheckingStatus	no_checking	17.54%
EmploymentDuration	greater_7	14.92%

XAI example: Loan approval dashboard

What features and how much they accounted for RISK (No Loan)

Confidence



Label is
"Risk"

XAI example: Loan approval dashboard

What features and how much they accounted for NO RISK (YES to Loan)

No Risk

Label is
"No Risk"

Confidence

44.56%

Factors contributing to **No Risk** confidence level

OthersOnLoan: none

10.21%

LoanDuration: 15

9.07%

Telephone: none

4.07%

Sex: male

3.70%

XAI example:

Loan approval dashboard

Minimum change
to flip the output

Maximum change
to keep the output

Minimum changes for No Risk outcome ⓘ

CreditHistory	outstanding_credit
LoanDuration	15
ExistingSavings	less_100

Maximum changes allowed for the same outcome ⓘ

LoanPurpose	retraining
Age	51.0
Job	management_self-employed

XAI example:

Loan approval dashboard

How this prediction was determined

The **GermanCreditRiskModel** predicts **Risk** with 55.44% confidence. The following features were most important in determining this prediction: Age (20.60%), CheckingStatus (17.54%), and EmploymentDuration (14.92%).

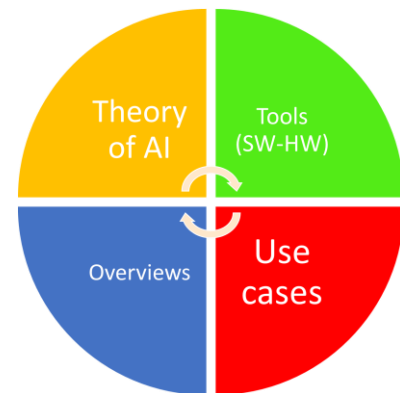
Text explanation of the decision



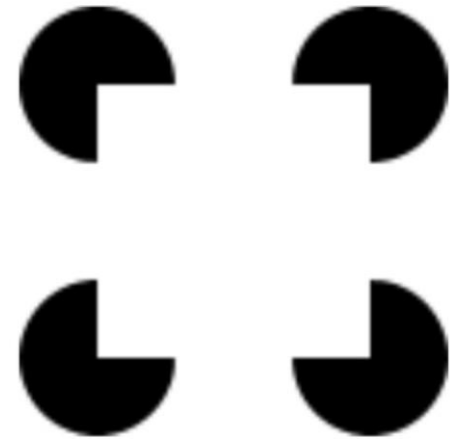
THEORY

The human Gestalt

Why we are so far to the human brain...



Human brain: Gestalt



- “Gestalt” (*): from a German word that roughly means *pattern* or *form*.
- “Whole is greater than the sum of its parts”
- “Learning ***is more than just invoking mechanical responses from learners***”.
- The experiences and perceptions of learners have a significant impact on the way that they learn.
- **This feature of the human brain is quite difficult to be mimicked from current AI models**

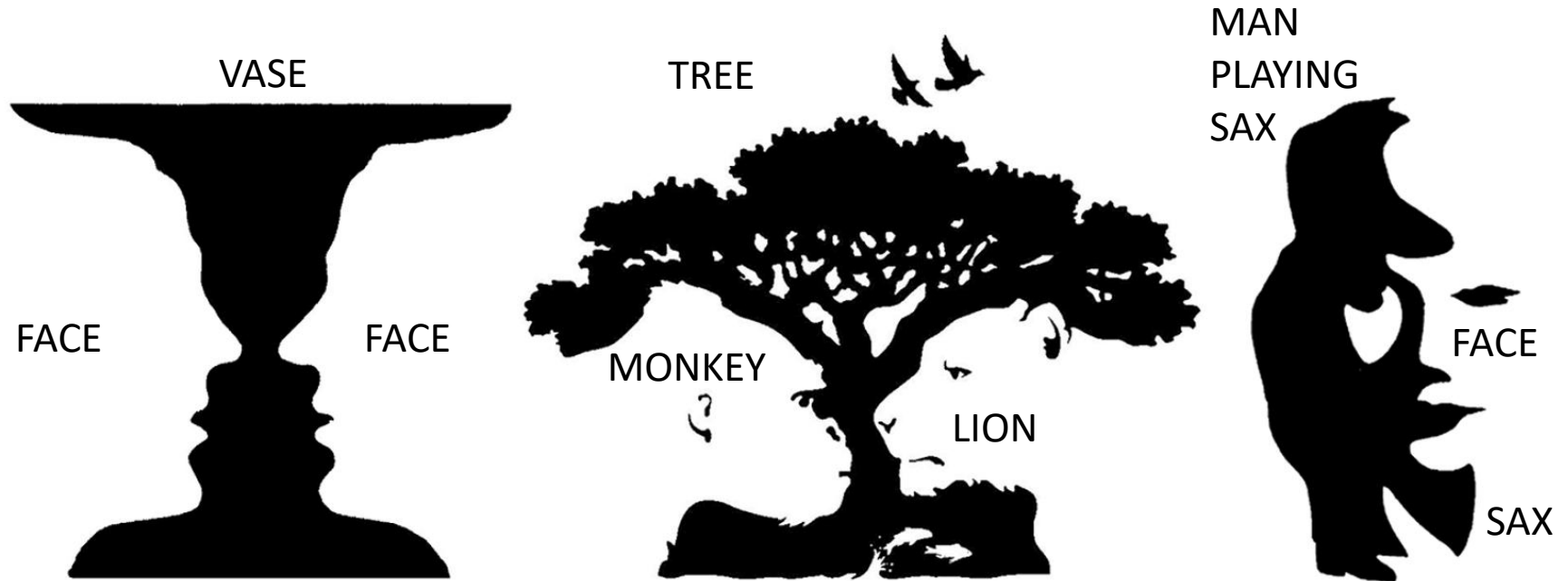
(*) from Graf Christian von Ehrenfels, Austrian philosopher (1859-1932)

Gestalt examples: foreground and background



A part of the image can be «just noise» for one pattern
or «the most salient feature» for the other

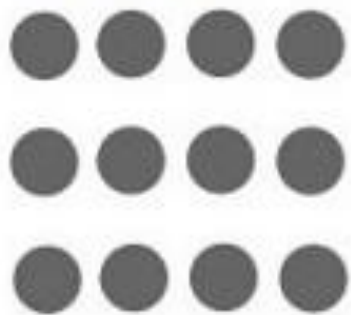
Gestalt examples: foreground and background



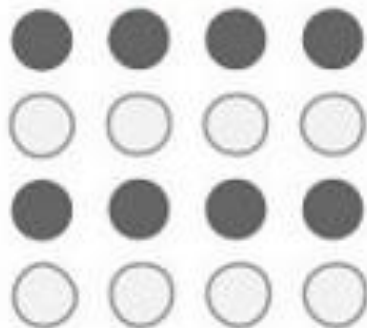
A part of the image can be «just noise» for one pattern
or «the most salient feature» for the other

Gestalt: laws of perceptual organization (examples)

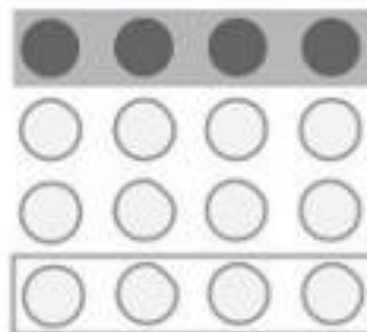
Proximity



Similarity



Enclosure



Symmetry



Closure



Continuity



Connection

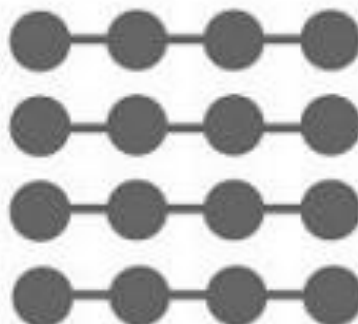
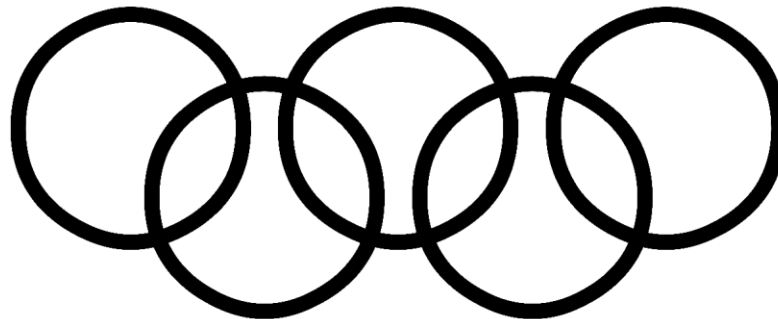


Figure & ground



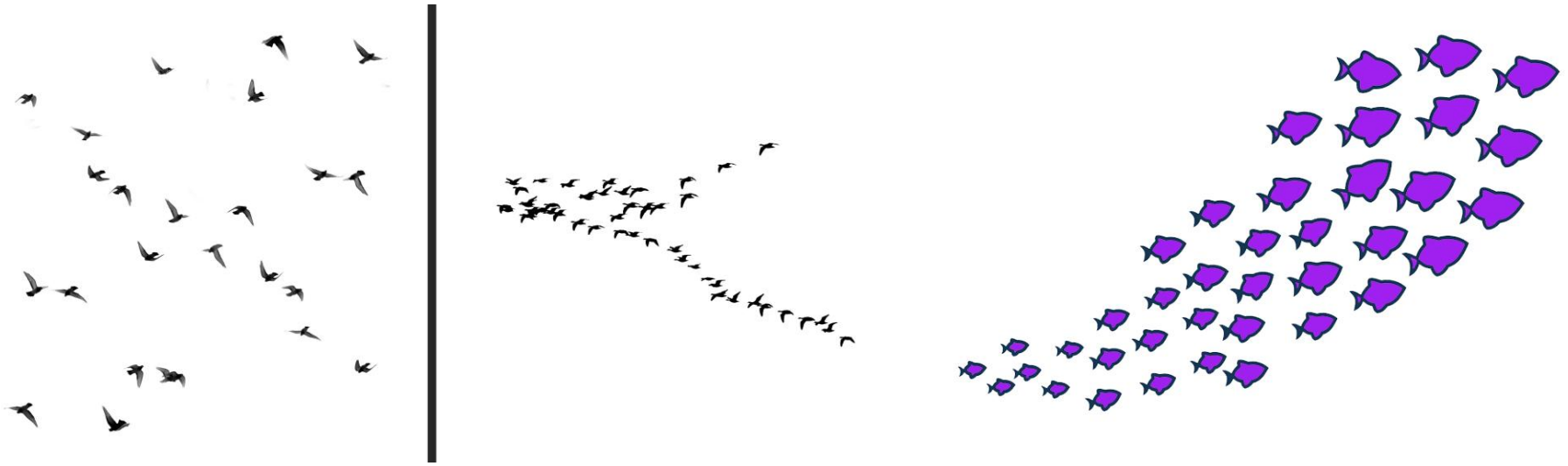
Gestalt: Law of Pragnanz ("good figure")

- That objects in the environment are seen in a way that **makes them appear as simple as possible**
- You see the image below as a series of overlapping circles rather than an assortment of curved, connected lines



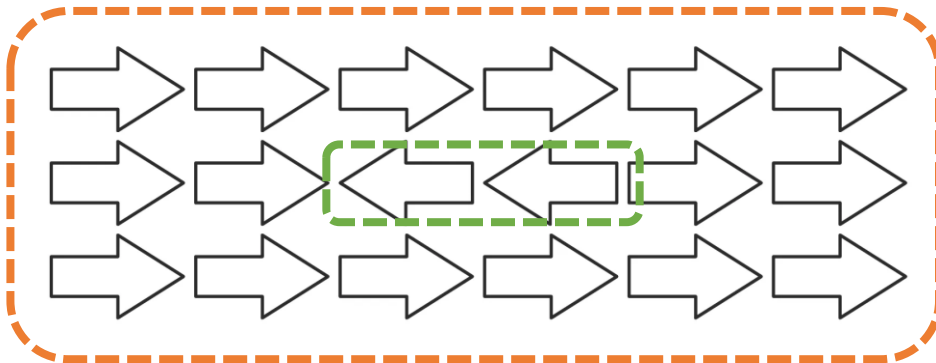
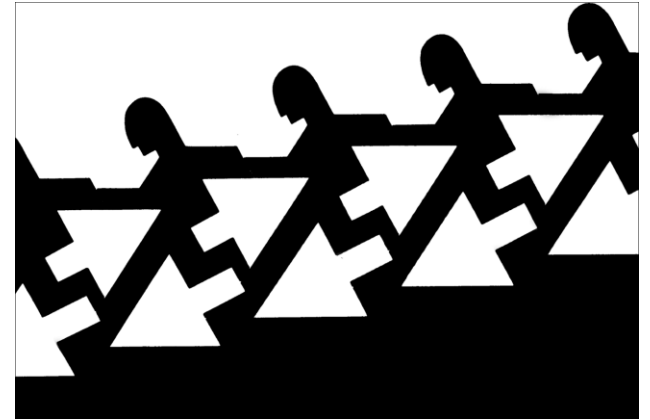
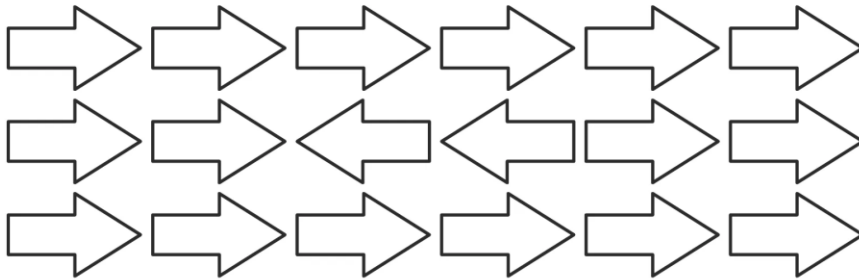
Gestalt: Common Fate

- humans perceive visual elements that move in the same speed and/or direction as parts of a single stimulus

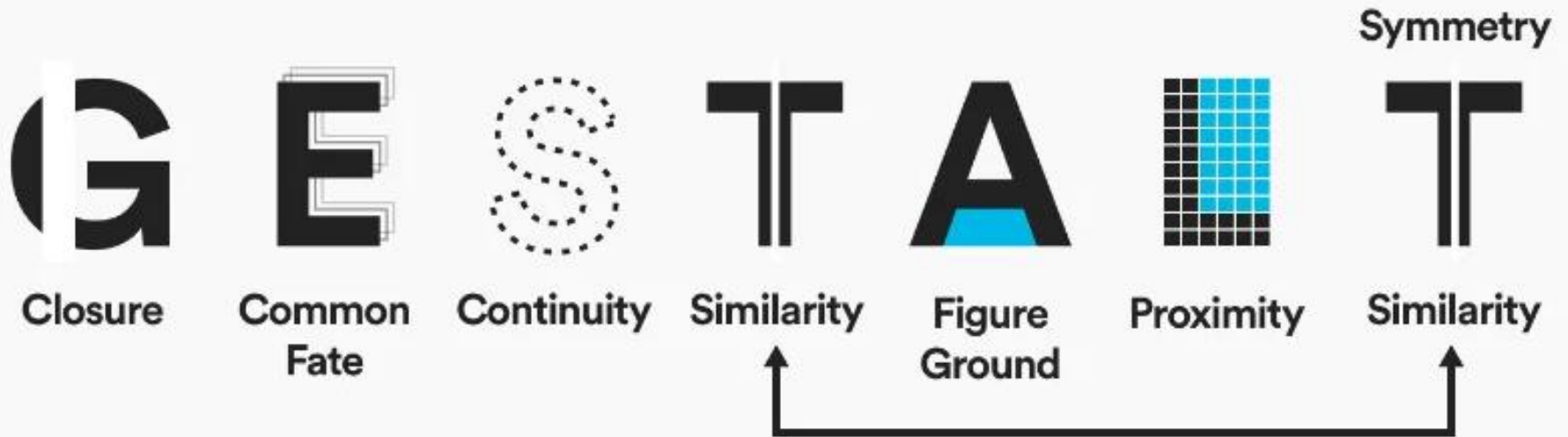


Gestalt: Common Fate

- same direction ...



Gestalt again...

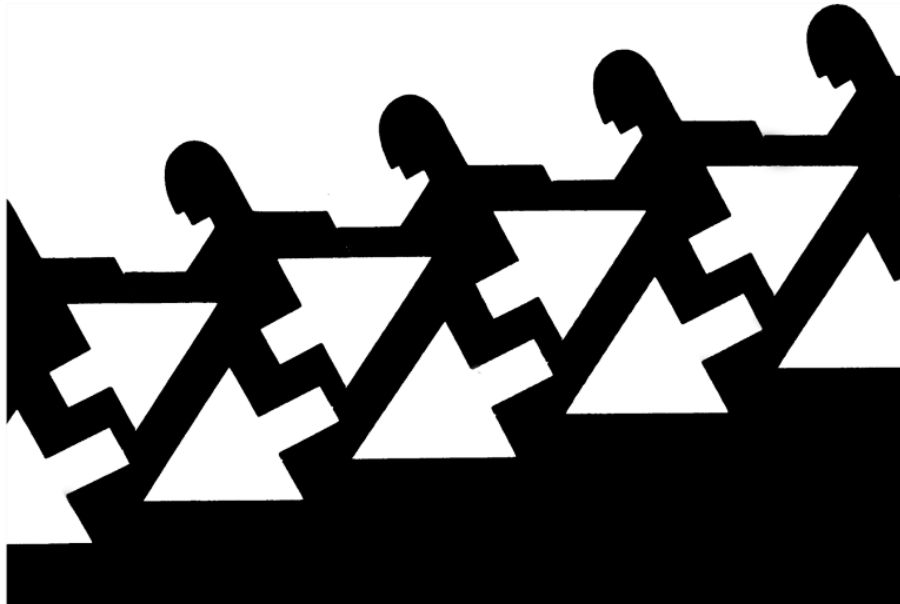


ChatGPT 4 attempt...

(Common fate + figure/ground)



You



what is it?

Inverting the background
...they are arrows



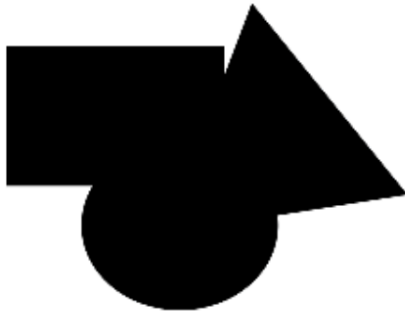
The image you've uploaded appears to be an abstract representation, featuring silhouetted **figures** that are repeated in a sequence, likely intended to depict motion or progression. The figures are simplified, **lacking detailed facial features**, and are placed against a zigzag pattern that could symbolize fluctuations, steps, or a progression of stages. **The zigzag base on which the figures are situated could also represent mountains or a graph with peaks and valleys.** The overall design is quite graphic and stylized, emphasizing form and rhythm over realism. It could be part of a visual identity for a brand, a graphic element for a presentation, or an artistic composition focusing on the theme of growth, development, or teamwork.

ChatGPT 4 attempt...

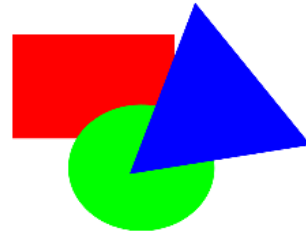
(Low of Pragnanz = “good figure.” in german)



You



what is it?



Nice job!



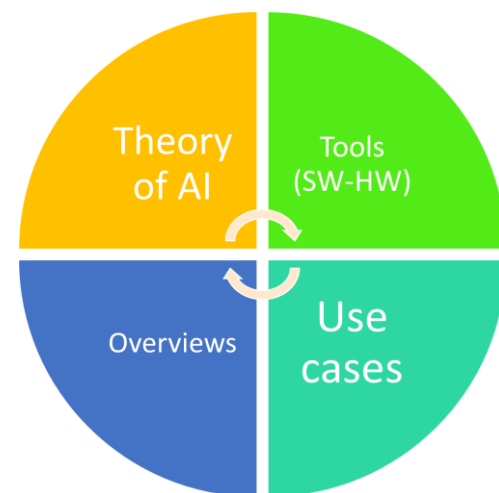
ChatGPT

The image you uploaded appears to be an abstract black symbol on a white background. It consists of a circle from which two shapes protrude: one is a horizontal rectangle, and the other is a triangle pointing to the right. This symbol does not correspond to any widely recognized iconography or notation in common use. It could be a logo, a stylized letter or number, or simply an abstract design. Without further context, it's difficult to provide a definitive answer about its meaning or purpose.



Overview

Opinions about AI in users and experts



Some recent data



The
Alan Turing
Institute

<https://www.adalovelaceinstitute.org/>

An independent research institute with a mission to ensure data and AI work for people and society

They published a nationally representative survey of public attitudes to artificial intelligence

November 2022, 4000 adults in Britain

https://www.turing.ac.uk/sites/default/files/2023-06/how_do_people_feel_about_ai_-_ada_turing.pdf

Figure 1: Awareness of uses of AI

'Before today, had you heard of the use of AI technologies for...'

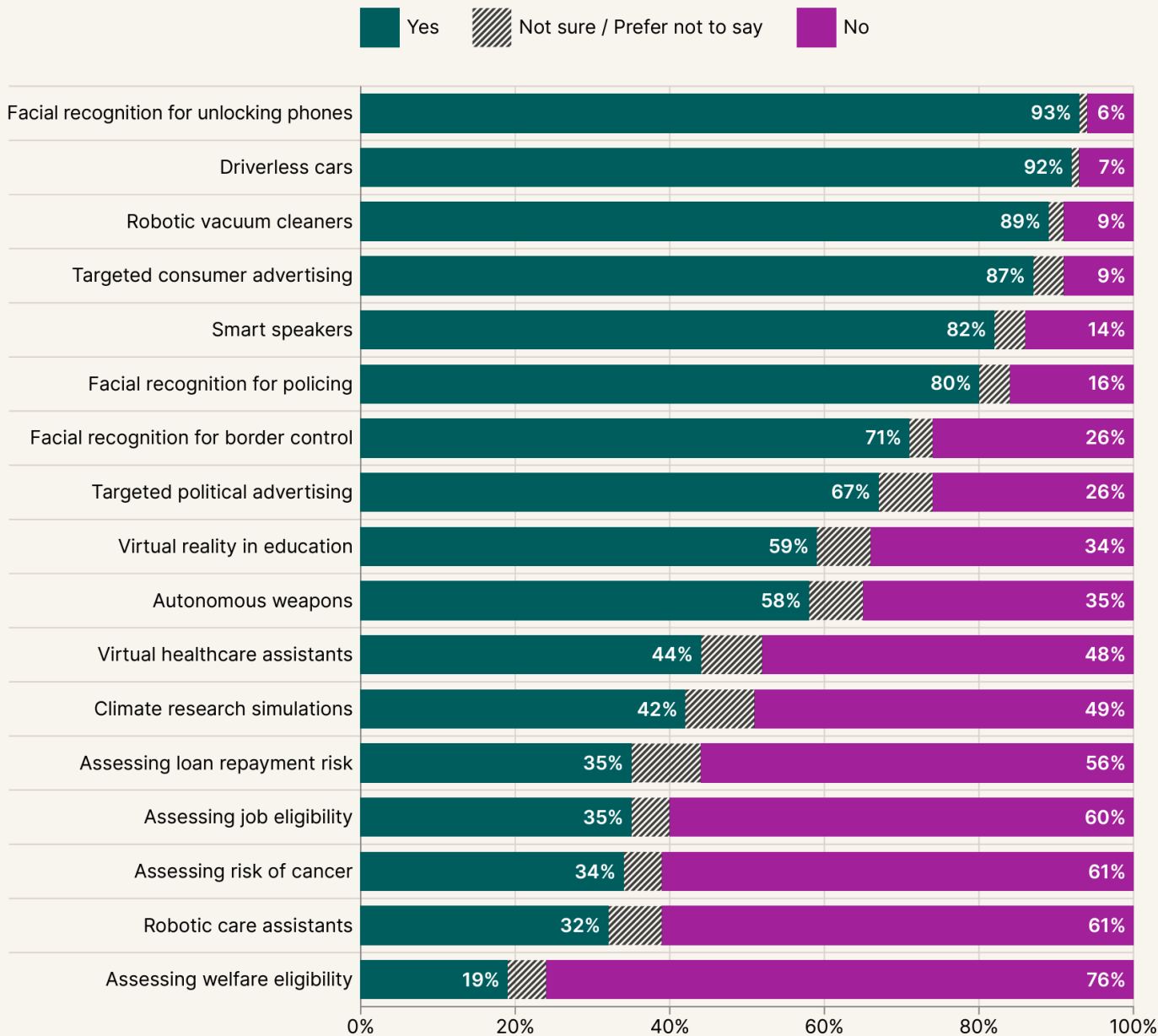


Figure 2: Experience with AI

'How much personal experience have you had with this technology?'

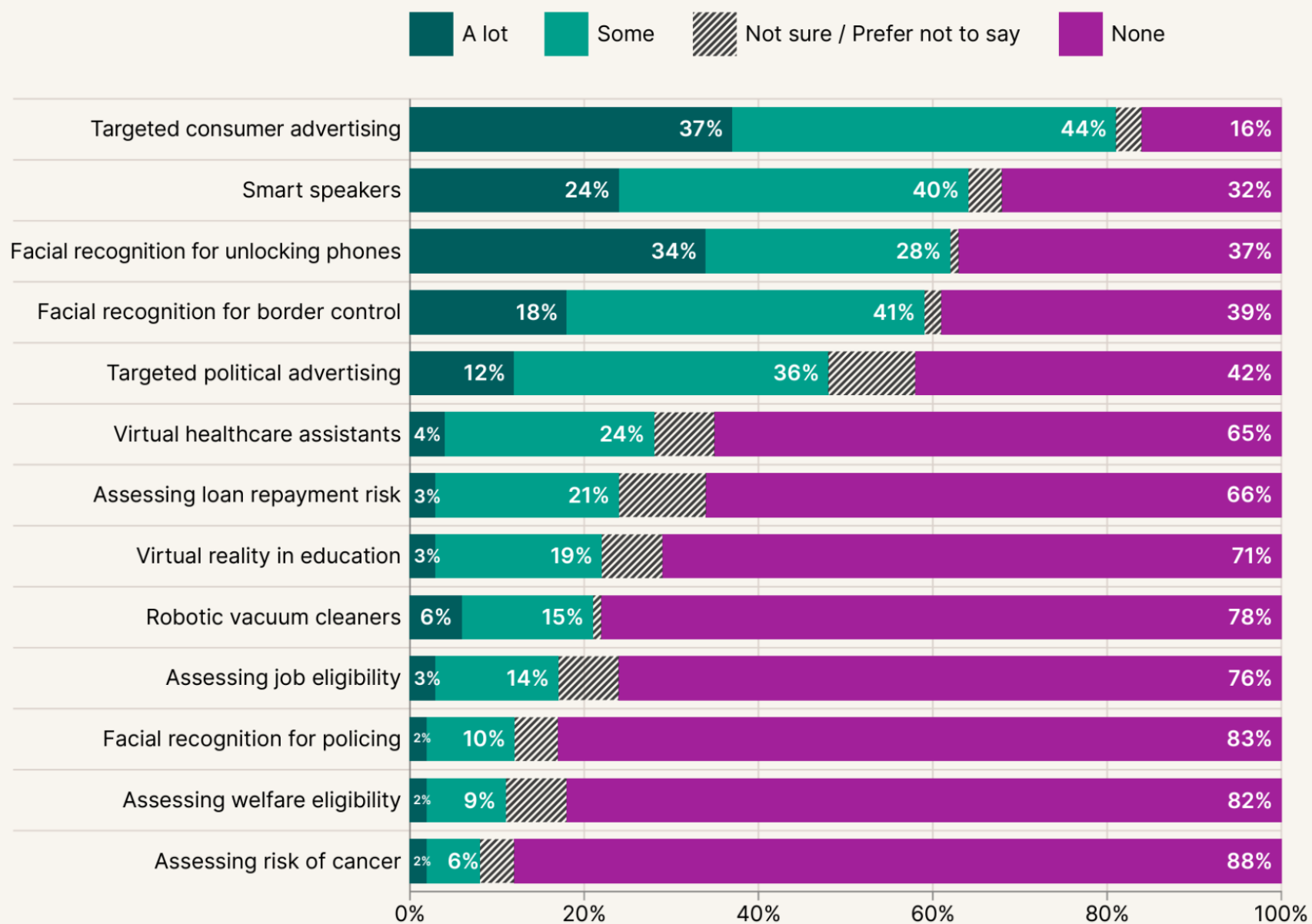
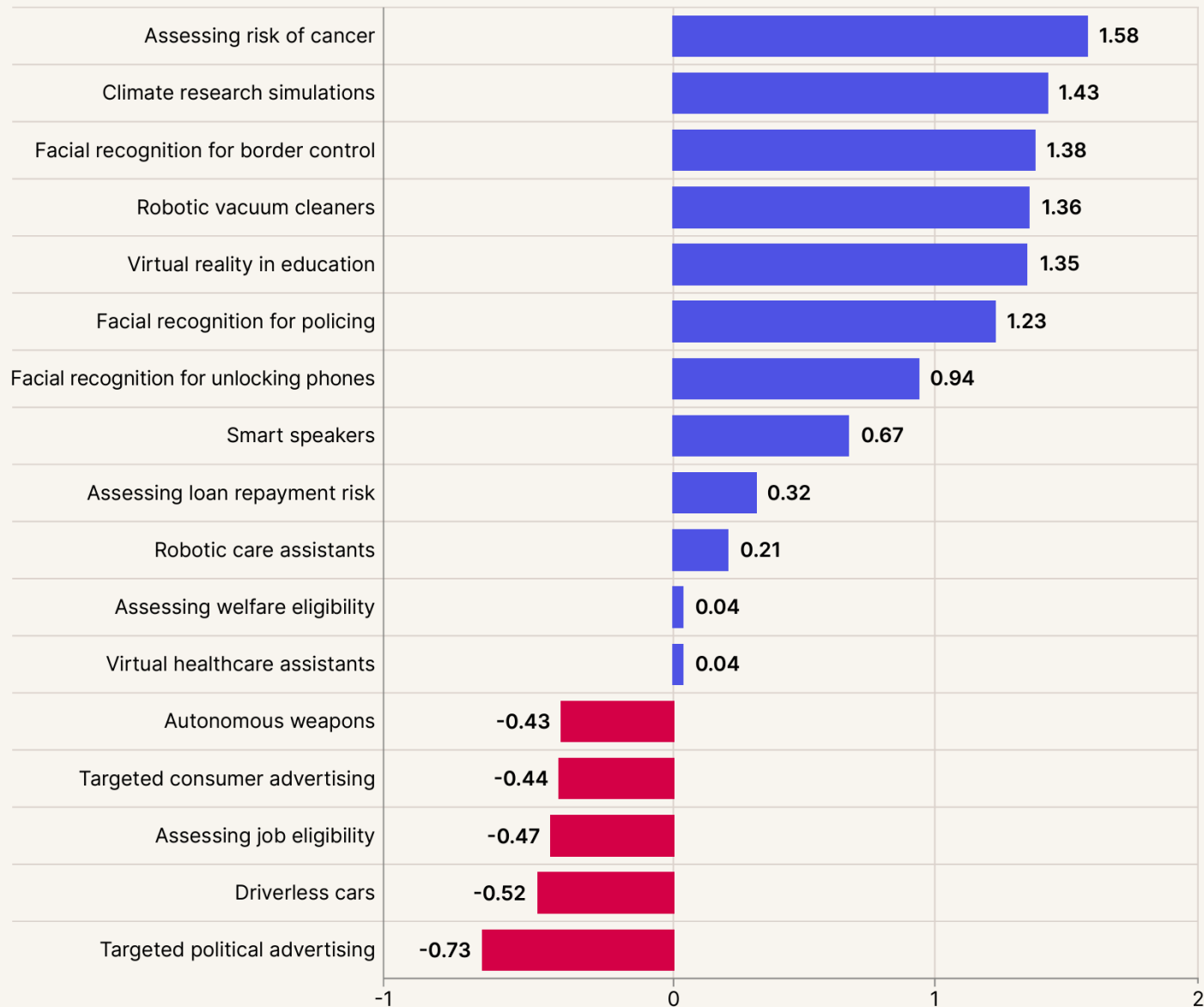


Figure 5: Net benefit score for each use of AI

Overall concern for each use of AI subtracted from overall perceptions of benefit (positive scores indicate that benefits outweigh concerns, while negative scores indicate that concerns outweigh benefits)



In summary

- For the majority of AI uses that we asked about, people had broadly positive views, but expressed concerns about some uses.
- Digging deeper into people's perceptions of AI shows that the British public hold highly nuanced views on the specific advantages and disadvantages associated with different uses of AI.
- The public want regulation of AI technologies, though this differs by age.

Some (positive) opinions

- “A **breakthrough** in machine learning would be worth ten Microsofts”, Bill Gates, Chairman, Microsoft
- “Machine learning is the **next Internet**”, Tony Tether, Director, DARPA
- Machine learning is the **hot new thing**”, John Hennessy, President, Stanford

Let's have a look also to different opinions (quite different opinions)...

Opinions: Stephen Hawking



- Artificial intelligence (AI) will be “either the best or worst thing” for humanity.
- “The development of full artificial intelligence could spell the end of the human race....
- It would take off on its own, and re-design itself at an ever-increasing rate.
- Humans, who are limited by slow biological evolution, couldn't compete, and would be superseded.”

Larry page



“Artificial intelligence would be the ultimate version of Google. The **ultimate search engine** that would **understand everything on the web**. It would understand exactly what you wanted, and it would give you the right thing. We're nowhere near doing that now. However, we can get incrementally closer to that, and that is basically what we work on.”

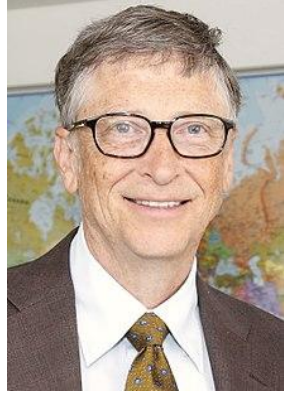
Elon musk

(probably changing opinion...)



- “The pace of progress in artificial intelligence (I’m not referring to narrow AI) is incredibly fast. Unless you have direct exposure to groups like Deepmind, you have no idea how fast—it is growing at a pace close to exponential. The risk of **something seriously dangerous happening** is in the five-year timeframe. **10 years at most.**”
- “I’m increasingly inclined to think that there should be some **regulatory oversight**, maybe at the national and international level, just to make sure that we don’t do something very foolish. I mean with artificial intelligence we’re summoning the demon.”

Bill Gates (recent opinion)



- “Humans should be worried about the threat posed by artificial intelligence.”
- “I am in the camp that is concerned about artificial intelligence. **First the machines will do a lot of jobs for us** and not be super intelligent. That should be positive if we manage it well. A few decades after that though the intelligence is strong enough **to be a concern**”.
- “I agree with Elon Musk and some others on this and don’t understand why some people are not concerned.”
→ Microsoft Invests \$10 Billion in ChatGPT Maker OpenAI
(Source: Bloomberg, 2023)

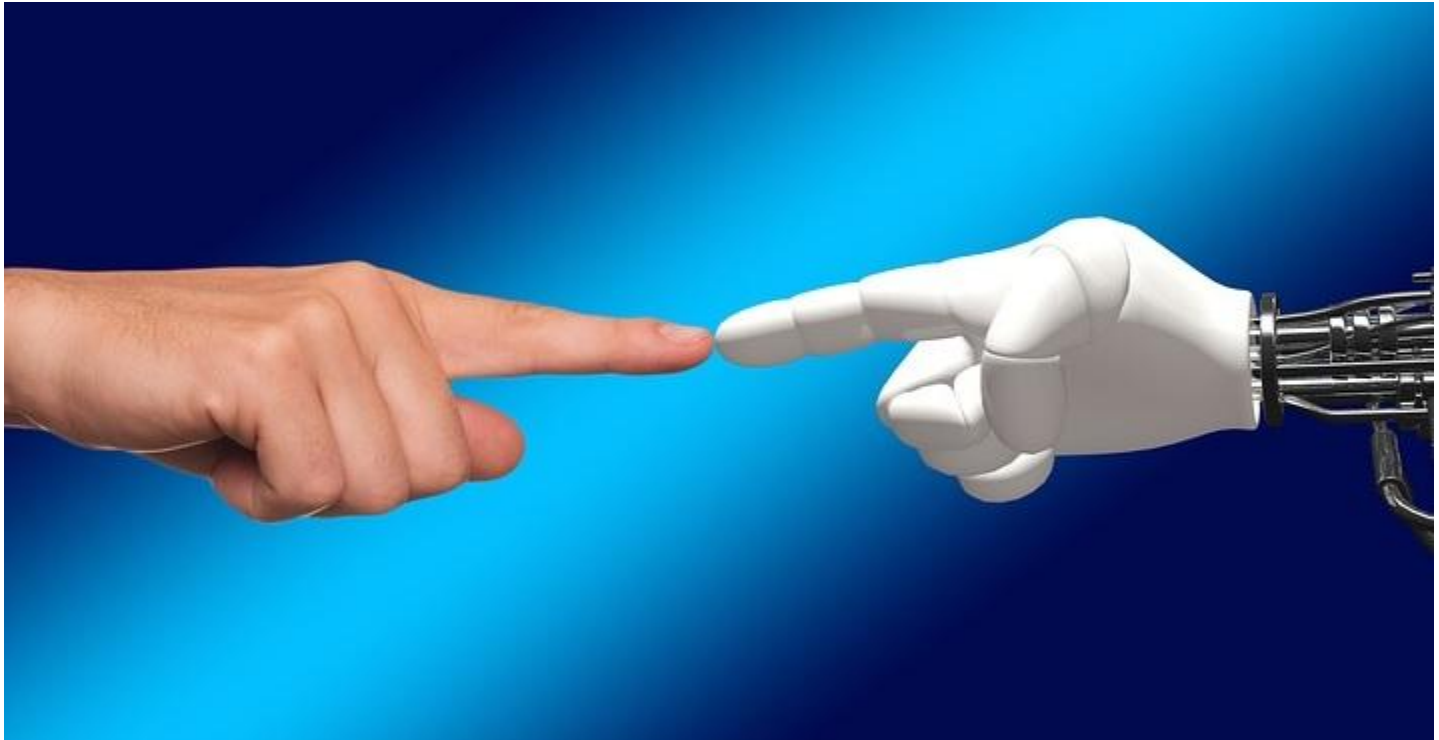
Andrew Ng

Computer scientist, investor, and writer
With >167000 citations....



- Worrying about AI evil superintelligence today is like worrying about overpopulation on the planet Mars. We haven't even landed on the planet yet!
- AI is the new Electricity

What is your opinion?

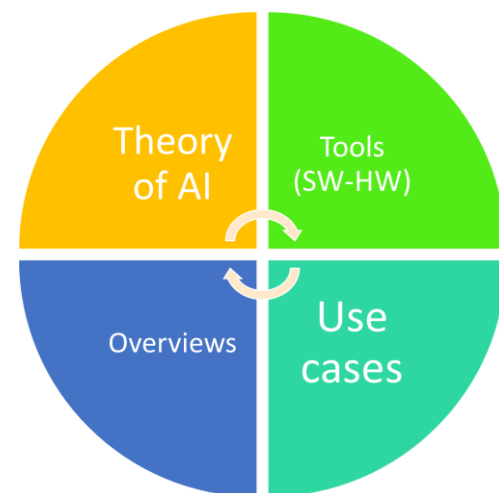


Write it down We'll go back to that

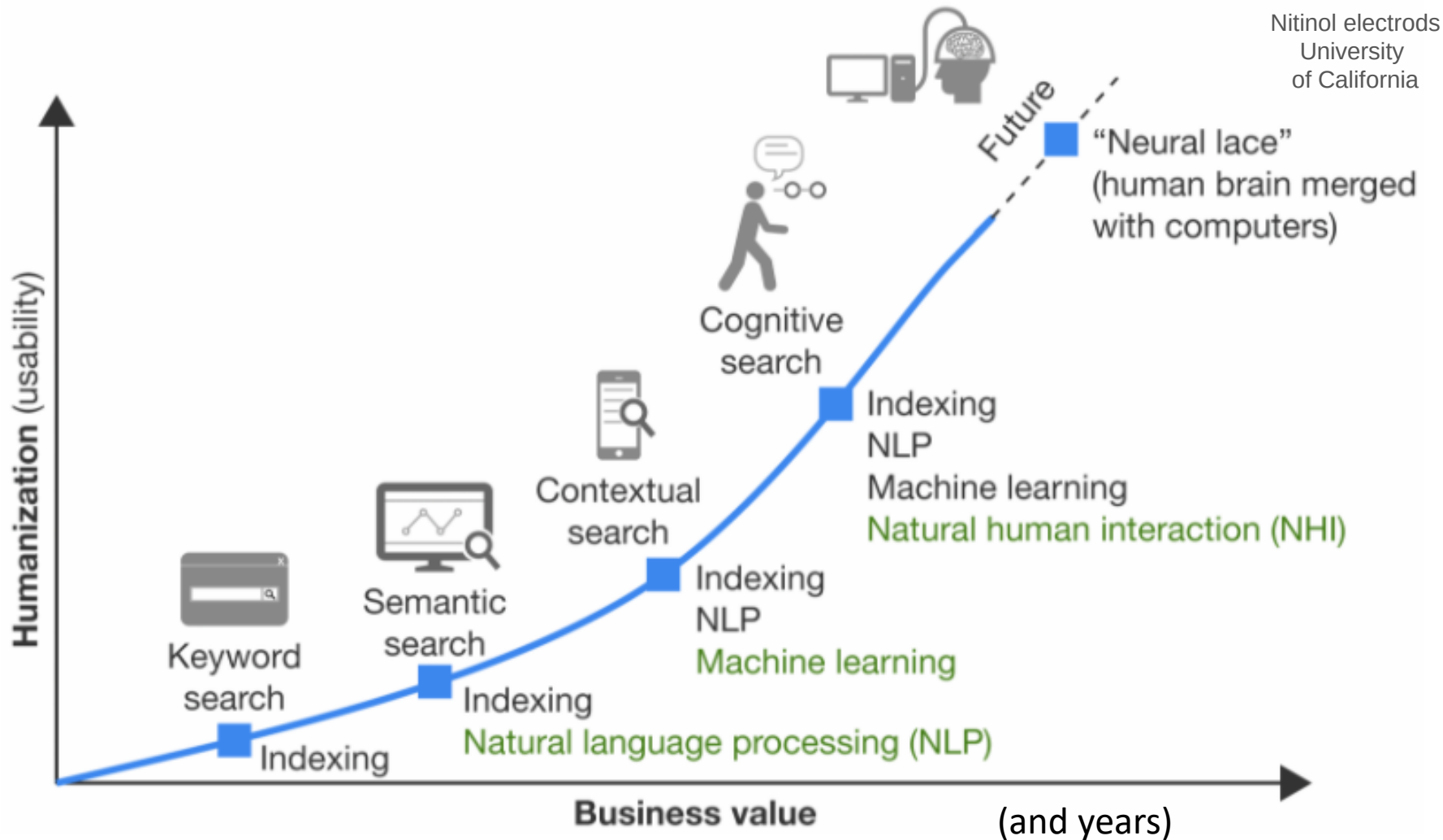


Overview

Usability and maturity

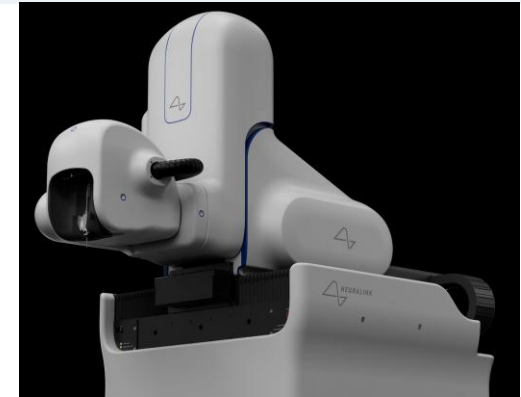
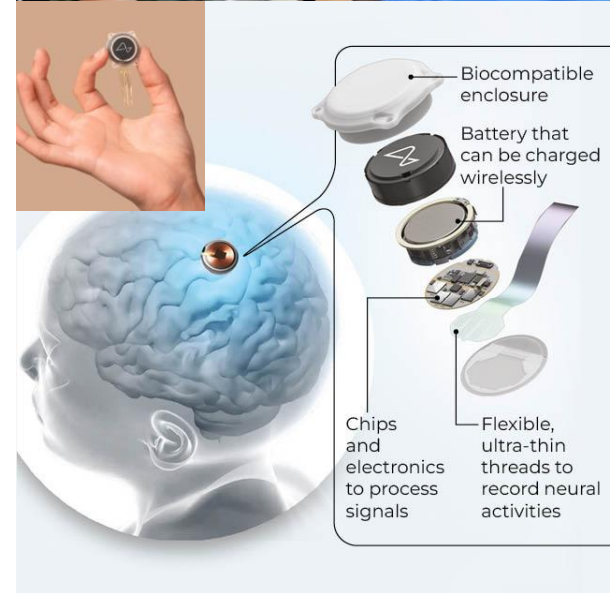


Usability in ML and Business value



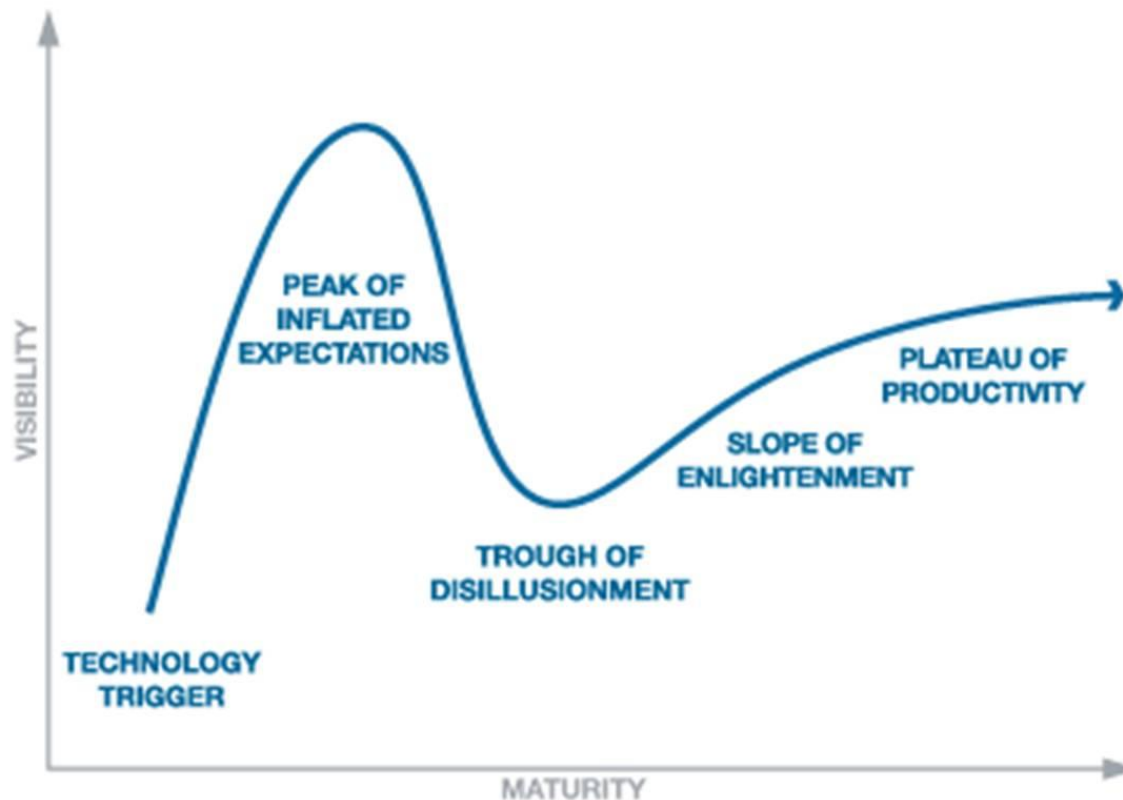
Example of advanced interaction

- Noland Arbaugh
 - First Neuralink patient to have the Telepathy chip implanted in his brain to allow him to use electronic devices solely with the power of thought.
 - became completely paralyzed from the shoulders down due to a diving accident eight years ago, now playing chess and Civilization as first step
- System: Battery, wireless charger, processor, I/O, electrodes, enclosure
- Neuralink uses a surgical robot to implant the ultra-thin wires with 1024 electrodes which are thinner than a human hair, in the specific regions of the brain where they need to be placed



The Gartner “Hype cycle”

(Expectations)



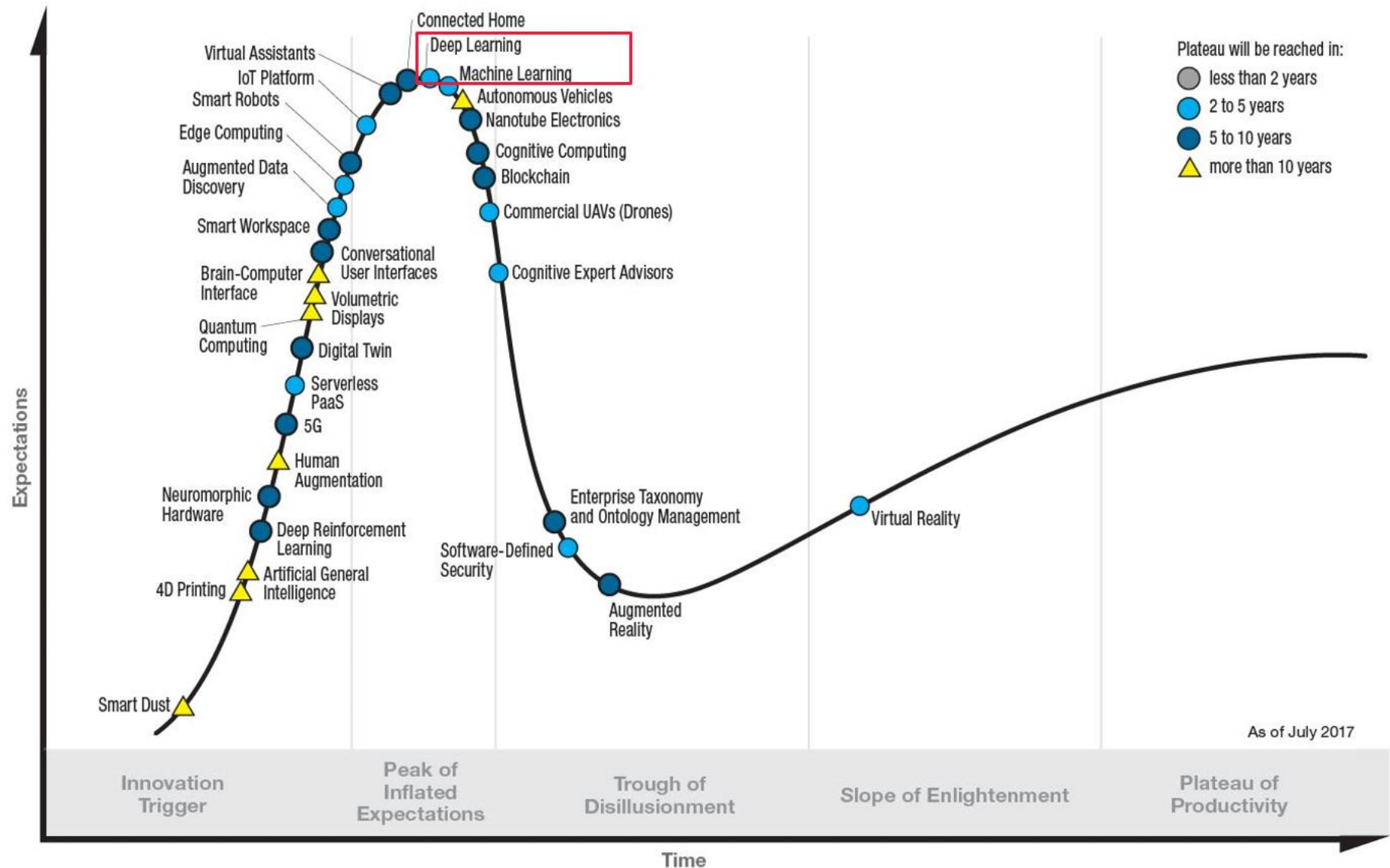
The plot represents the maturity, adoption, and social application of specific technologies.

Ok **but**...

- is not a cycle
- is not scientific in nature,
- no changes over time in the speed at which technology develops
- ... and many other
- But's is interesting...

Gartner: American research, advisory and information technology firm

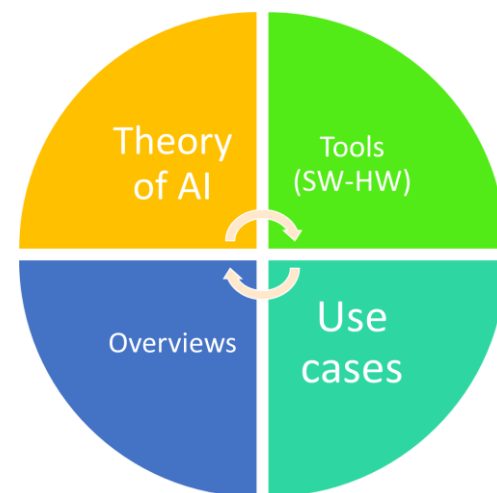
The AI/ML hype cycle graph





Toolboxes

The 4 main drivers

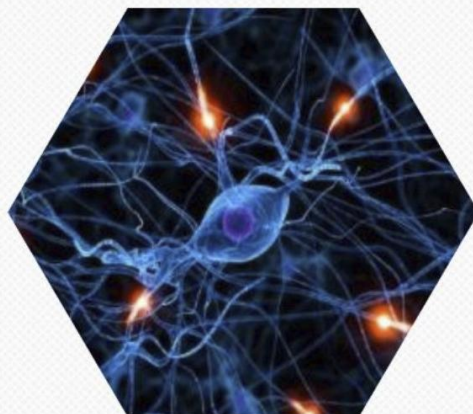


The three main (classical) drivers for AI

More Data



Better Models and Algorithms



Powerful GPU Accelerators



Keep it in mind...

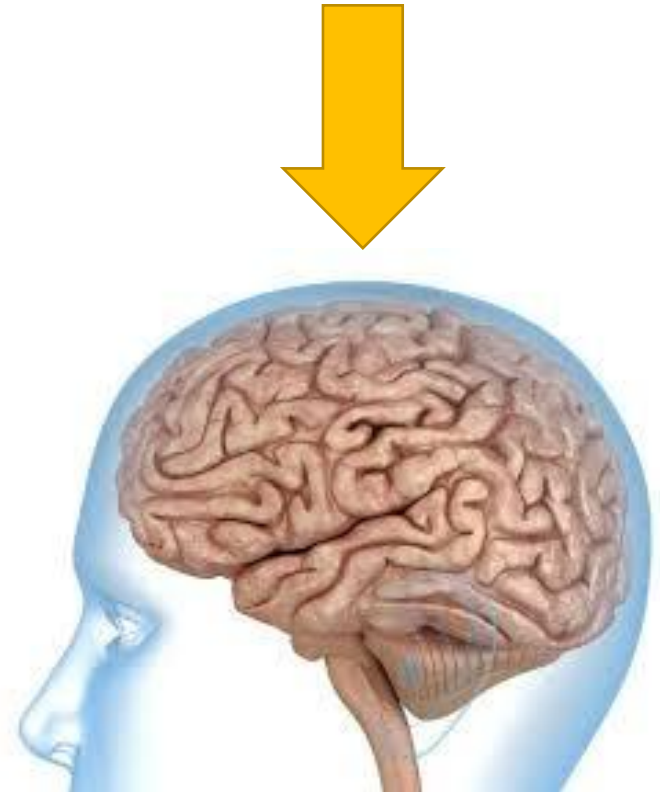


Especially for Deep Learn

The 4° driver

- The designer's brain!
 - A priori knowledge
 - Data selection
 - Data Filtering and Enhancing
 - Model selection
 - Learning technique choice
 - Experiment Design
 - Avoid brute force
 - Improve Explainability
 - Hybrid systems
 - Dividi et impera

Intelligent systems for industry,
supply chain and environment



Main points



- AI is now almost everywhere (hype) and still improving but
 - It is not known what “exactly” a n. network is learning
 - Some relevant human features are not (yet) available
- Small perturbation of the inputs can provide a large change in the output of a neural network
 - **Again:** it is not clear what the network is learning
- Explainable AI (XAI)
- Gestalt
- Opinions about AI are quite different
- The 4th driver (the designer’s brain) is the most important one!